Introductory Abstract Algebra
A Course Note

Daniel Le

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Preface

These notes are maintained by Daniel Le as course notes for his *Abstract Algebra (MA333)* class. The notes mainly are derived from *A First Course in Abstract Algebra* by Joseph Rotman [1]. Of course, it is not perfect and is being reviewed by Daniel on a regular basis. If you are anyone but Daniel and reading this, please do it with both a critical and accepting mind. I can make mistakes, especially in the *Exercises* subsections where the answers are all noone's but mine.

I keep these notes mainly in my own interest when I want to do reviews for the course. That is why in some sections I will write something along the lines of "left for the readers". Keep in mind when you see those lines: Daniel understood all the steps being omitted; he is just too lazy to fill in the details. You should be able to fill in the blank yourself. Some of the proofs are written in a very concise way. Nothing is missing but some small details, which, again, are personally understood by Daniel. If you feel like a transition from one step to another is too swift, try to work out all the steps of the argument. If an argument makes no sense whatsoever, I *absolutely can* be making a mistake at that point.

The sections in this notebook are mostly small sections in big Chapters in the books, which is why I call them Sections instead of Chapters. Personally, I feel like things are more organized this way. This does mean, for instance, *symmetric groups* will be mentioned in the Geometry section and not the Groups section. Things like this will happen in the notes: if something clearly very important is mentioned in the book but not in these notes, you are just probably looking at the wrong place. Only a small selection of contents are not included, which will be listed right below. At this point, 7/2/2026, these notes are **NOT** finished. Expect more updates to come.

These course notes should not be used as a substitute for a textbook, since everything has an answer key. I also only included problems that personally appear to be "hard" or important results. You should definitely look in the books: there are a lot of good problems there that I might have skipped over. In these notes, I did **NOT** include a majority of results from the topics of *Number Theory*, *Equivalence Classes*, *Functions*, and *Linear Transformations*. In order to learn these topics, I recommend these Colby courses: *Linear Algebra (MA253)*, *Mathematical Reasoning (MA274)* or the *Honors Calculus (MA135-165)* sequence, and *Elementary Number Theory (MA357)*. Of course, I got the privilege to have taken these courses before these notes were created. Despite this, some results I find particularly interesting but are not mentioned in the main sections may be mentioned in the Appendix.

If you are still reading, that is all I have to disclaim. I am always available through email at dtle28@colby.edu or danieltle0602@gmail.com. Please shoot me an email if you think I am violently wrong or lacking in any parts of these notes. If you are cool enough to be my friend, shoot me a message. I love me some heated math discussions.

Alrighty, that is all. Enjoy!

Daniel Le

Contents

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Permutations

Definitions

Definition 1.1 (Permutations)**.** A *permutation* of a set X is a bijective function $f : X \to X$.

Definition 1.2 (Notation)**.** A rearrangement determines the function $\alpha : X \to X$ defined as $\alpha(j) = i_j$. We use a tow-rowed notation to denote the function corresponding to a rearrangement. If $\alpha(j)$ is the j−th item on the list, then

$$\alpha = \begin{pmatrix} 1 & 2 & \dots & n \\ \alpha(1) & \alpha(2) & \dots & \alpha(n) \end{pmatrix}.$$

Definition 1.3 (Symmetric Group)**.** The family of all permutations of a set X, denoted by S_X, is called the *symmetric group* on X. When $X = \{1, 2, \dots, n\}$, S_X is usually denoted by S_n, and it is called the *symmetric group on n letters*.

Remark 1.4. Composition in S_X satisfies the *cancellation law*: If $\gamma \circ \alpha = \gamma \circ \beta$ then $\alpha = \beta$. The same conclusion applies when $\alpha \circ \gamma = \beta \circ \gamma$

Definition 1.5. If $\alpha \in S_n$ and $i \in \{1, 2, \dots, n\}$, then α *fixes* i if $\alpha(i) = i$, and α *moves* i is $\alpha(i) \neq i$.

Definition 1.6. Let $i_1, i_2, \dots, i_r$ be distinct integers in $\{1, 2, \dots, n\}$. If $\alpha \in S_n$ fixes the other integers (if any) and if

$$\alpha(i_1) = i_2, \alpha(i_2) = i_3, \dots, \alpha(i_{r-1}) = i_r, \alpha(i_r) = i_1,$$

then α is called an *r-cycle*. One also says that α is a *cycle* of *length* r.

Definition 1.7 (Cycle Notation)**.** An *r-cycle* shall be denoted by $\alpha = (i_1\ i_2\ \dots\ i_n)$.

Definition 1.8 (Disjoint Permutations)**.** Two permutations $\alpha, \beta \in S_n$ are *disjoint* is every i is moved by one is fixed by the other: if $\alpha(i) \neq i$, then $\beta(i) = i$, and if $\beta(j) \neq j$, then $\alpha(j) = j$. A family $\beta_1, \dots, \beta_r$ of permutations is *disjoint* if each pair of them is disjoint.

Remark 1.9. Two cycles $\alpha = (i_1\ i_2\ \dots\ i_r)$ and $\beta = (j_1\ j_2\ \dots\ j_s)$ are disjoint if and only if $\{i_1, i_2, \dots, i_r\} \cap \{j_1, j_2, \dots, j_s\} = \varnothing$. When α and β are disjoint, there are only three possibilities for a number i: moved by α, moved by β, or moved by neither.

Definition 1.10 (Complete Factorization)**.** A *complete factorization* of a permutation α is a factorization of α into disjoint cycles that contains one 1-cycle (i) for every i fixed by α.

Definition 1.11. A permutation $\alpha \in S_n$ is *even* if it can be factored into a product of an even number of transpositions; otherwise, α is *odd*.

Definition 1.12 (Alternating Group)**.** The subset of all even permutations is called the *alternating group*, denoted by A_n. Later in the course notes, we will know this is a *subgroup* of S_n.

Definition 1.13. If $\alpha \in S_n$ and $\alpha = \beta_1\beta_2 \dots \beta_t$ is a complete factorization into disjoint cycles, then *signum* α is defined by $\operatorname{sgn}(a) = (-1)^{n-t}$.

Remark 1.14. Notice that $\operatorname{sgn}(\varepsilon) = 1$ for every 1-cycle ε because $t = n$. If τ is a transposition, then it moves two numbers, say, i and j, and fixes the other $n-2$ numbers. Therefore, $t = (n-2)+1 = n-1$: there are $(n-2)$ 1-cycles and 1 transposition, all disjoint. It follows that $\operatorname{sgn}(\tau) = (-1)^{n-(n-1)} = -1$.

Theorems

Theorem 1.15. *Every permutation $\alpha \in S_n$ is either a cycle or a product of disjoint cycles.*

Proof. We use complete induction on k, the number of items moved by α, for this proof. We also denote α_k as the permutation that moves k terms. For the base case where $k = 0$, the statement is true since α_0 is the identity, and is a 1-cycle. Suppose that for all $0 \le m \le k-1$, α_m is a cycle or a product of disjoint cycles. We show α_k is also a cycle or product of disjoint cycles. Let i_1 be a point moved by α_k. Define $i_2 = \alpha_k(i_1), i_3 = \alpha_k(i_2), \dots, i_{r+1} = \alpha_k(i_r)$, where $r = \min\{m \in \mathbb{N} : i_{m+1} \in \{i_1, i_2, \dots, i_m\}\}$. We claim that $\alpha_k(i_r) = i_1$. Otherwise, $\alpha_k(i_r) = \alpha_k(i_j)$ for some $j \ge 2$; but $\alpha_k(i_{j-1}) = i_j$ also. This contradicts the injectivity of α. Define $\sigma = (i_1\ i_2\ \dots\ i_r)$. If $k = r$, then $\alpha_k = \sigma$ is a cycle. If $r < k$, let Y be the remaining $k-r$ points. By their definitions, we have $\alpha_k(Y) = Y$ while σ fixes the points in Y and the restrictions $\alpha|_{\{i_1,i_2,\dots,i_r\}} = \sigma|_{\{i_1,i_2,\dots,i_r\}}$. Let α' be the permutation such that $\alpha'|_{\{i_1,i_2,\dots,i_r\}}$ fixes $i_1, i_2, \dots, i_r$ and $\alpha'|_Y = \alpha|_Y$. We know that σ and α' are disjoint and that $\alpha = \sigma\alpha'$. Furthermore, since α' moves fewer points than α, it is also a cycle or product of disjoint cycles. Either way, α is a product of disjoint cycles, which completes the proof. ⌑

Theorem 1.16. *Let $\alpha \in S_n$ and let $\alpha = \beta_1\beta_2 \dots \beta_t$ be a complete factorization into disjoint cycles. This factorization is unique up to orders of cycles.*

Proof. Since every complete factorization of α has exactly one 1-cycle for each i fixed by α, $t = 1$ exactly when $n = 1$ and $\alpha = \beta_1$, which is unique. For $k \ge 2$, suppose $\alpha = \gamma_1\gamma_2 \dots \gamma_s$ be a second, of course different, complete factorization of α into disjoint cycles. If β_t moves i_1, then $\beta_t^k(i_1) = \alpha^k(i_1)$ for all $k \ge 1$, by Lemma 1.23. Now, some γ_j must move i_1; since disjoint cycles commute, assume γ_s moves i_1, and, as above, $\gamma_s^k(i_1) = \alpha^k(i_1)$ for all k. It follows from Lemma 1.23 that $\beta_t = \gamma_s$, and the cancellation law yiels $\beta_1 \dots \beta_{t-1} = \gamma_1 \dots \gamma_{s-1}$. The proof is completed by induction on $\max\{t, s\}$. ⌑

Theorem 1.17. *The following statements hold true:*

(i) The inverse of $(i_1\ i_2\ \dots\ i_r)$ *is the r-cycle* $(i_r\ i_{r-1}\ \dots\ i_1)$.

(ii) If $\alpha = \beta_1\beta_2\dots\beta_t$ *is a product of disjoint cycles, then* $\alpha^{-1} = \beta_1^{-1}\beta_2^{-1}\dots\beta_r^{-1}$.

Proof. Here are some proof sketches:

(i) If first cycle maps $i_k \mapsto i_{k+1}$, the second cycle maps $i_{k+1} \mapsto i_k$.

(ii) It is clear that $\alpha^{-1} = \beta_r^{-1}\beta_{r-1}^{-1}\dots\beta_1^{-1}$, and since disjoint cycles commute, $\alpha^{-1} = \beta_1^{-1}\beta_2^{-1}\dots\beta_r^{-1}$.

And the proofs are complete. ⌑

Theorem 1.18. *If* $n \geq 2$, *then every* $\alpha \in S_n$ *is a product of transpositions.*

Proof. By Theorem 1.15, it suffices to factor an $r-$cycle into a product of transpositions. This is done by factoring $(1\ 2\ \dots\ r) = (1\ r)\,(1\ r-1)\,,\dots,(1\ 2)$. ⌑

Theorem 1.19. *For all* $\alpha, \beta \in S_n$, $\operatorname{sgn}(\alpha\beta) = \operatorname{sgn}(\alpha)\operatorname{sgn}(\beta)$. *We say* sgn *is multiplicative.*

Proof. Assume that $\alpha \in S_n$ and $\alpha = \tau_1\tau_2\dots\tau_m$ is a factorization of α into transpositions with m minimal, which exists by Theorem 1.18. We prove the theorem by induction on m. For the base case where $m = 1$, the theorem is true by Lemma 1.25. Suppose for $m = k-1 \geq 1$, the statement is true. We show it is true for $m = k$. Notice that the factorization $\tau_2\dots\tau_k$ is also minimal; otherwise, $\tau_2\dots\tau_k = \sigma_2\dots\sigma_q$ where $q < k-1$, which makes $\alpha = \tau_1\sigma_2\dots\sigma_q$, contradicting the minimality of k. Therefore, $\operatorname{sgn}(\alpha\beta) = \operatorname{sgn}(\tau_1\tau_2\dots\tau_m\beta) = -\operatorname{sgn}(\tau_2\dots\tau_m\beta)$ by Lemma 1.25. By the induction hypothesis, $\operatorname{sgn}(\alpha\beta) = -\operatorname{sgn}(\tau_2\dots\tau_m)\operatorname{sgn}(\beta)$. By Lemma 1.25, $\operatorname{sgn}(\alpha\beta) = \operatorname{sgn}(\tau_1\tau_2\dots\tau_m)\operatorname{sgn}(\beta) = \operatorname{sgn}(\alpha)\operatorname{sgn}(\beta)$. ⌑

Theorem 1.20. *The following statements hold true:*

(i) A permutation $\alpha \in S_n$ *is even if* $\operatorname{sgn}(\alpha) = 1$ *and is odd if* $\operatorname{sgn}(\alpha) = -1$.

(ii) A permutation $\alpha \in S_n$ *is odd if and only if it is a product of an odd number of transpositions.*

Proof. Here are the proofs:

(i) Let $\alpha = \tau_1\tau_2\dots\tau_m$ be a factorization of α into transpositions with minimal m. By Theorem 1.19, $\operatorname{sgn}(\alpha) = \operatorname{sgn}(\tau_1)\operatorname{sgn}(\tau_2)\dots\operatorname{sgn}(\tau_m) = (-1)^m$. Thus, if $\operatorname{sgn}(\alpha) = 1$, m must be even, and if $\operatorname{sgn}(\alpha) = -1$, m must be odd.

(ii) Let $\alpha = \tau_1\tau_2\dots\tau_m$ be a factorization of α into transpositions with minimal m. We have established above that if m is odd, $\operatorname{sgn}(\alpha) = -1$, and α is odd. Conversely, if m is even, $\operatorname{sgn}(\alpha) = 1$ and α is even, which proves the contrapositive.

And the proofs are done. ⌑

Theorem 1.21. *For any $n \geq 3$, any $\alpha \in A_n$ is a 3-cycle or a product of 3-cycles.*

Proof. Since each even permutation can be factored an even number of transpositions, it suffices to show that a pair of transpositions can be written as a product of 3-cycles. If the two transpositions are identical, $(a\ b)\,(a\ b) = \mathbb{1}$, and this can be written as a product of a 3-cycle and its inverse. If they share 1 common element, we know $(a\ b)\,(b\ c) = (a\ b\ c)$ is a 3-cycle. If they are completely disjoint, let them be $(a\ b)$ and $(c\ d)$, we can introduce a shared element $(a\ b)\,(b\ c)\,(b\ c)\,(c\ d) = (a\ b\ c)\,(b\ c\ d)$ is a product of 3-cycles. ⌑

Lemmas

Lemma 1.22. *Disjoint permutations $\alpha, \beta \in S_n$ commute.*

Proof. It suffices to show that if $1 \leq i \leq n$, then $\alpha\beta(i) = \beta\alpha(i)$. If i is moved by β, say, $\beta(i) = j \neq i$, then β moves j as well; otherwise, $\beta(j) = \beta(i) = j$, which contradicts the injectivity of β. Since α and β are disjoint, $\alpha(i) = i$ and $\alpha(j) = j$. Hence, $\beta\alpha(i) = j = \alpha\beta(j)$. The same argument applies when i is moved by α. The only other case is when i is moved by neither α or β, which then implies $\alpha\beta(i) = i = \beta\alpha(i)$. Therefore, $\alpha\beta = \beta\alpha$. ⌑

Lemma 1.23. *The following statements hold true:*

(i) Let $\alpha = \beta_1\beta_2 \ldots \beta_t$ be a factorization of α. If β_1 moves i, then $\alpha^k(i) = \beta_1^k(i)$ for all $k \geq 1$.
(ii) If β and γ are cycles both of which move some i, and if $\beta^k(i) = \gamma^k(i)$ for all $k \geq 1$, then $\beta = \gamma$.

Proof. Here are the proofs:

(i) Denoting $\beta_2 \ldots \beta_t = \sigma$, we have $\alpha = \beta_1\sigma$. Since σ and β are disjoint and β_1 moves i, $\sigma(i) = i$, which implies $\sigma^k(i) = i$ as well. By Lemma 1.22, they commute, so $(\beta_1\sigma)^k = \beta_1^k\sigma^k$, according to Exercise 1.1. Hence $a^k(i) = (\beta_1\sigma)^k(i) = \beta_1^k(i)\sigma^k(i) = \beta_1^k(i)$.
(ii) Denote i by i_1. If $\beta = (i_1\ i_2\ \ldots\ i_r)$, then $i_2 = \beta(i_1)$, $i_3 = \beta(i_2) = \beta^2(i_1)$, and so on. We can inductively define $i_{k+1} = \beta^k(i_1)$ for all $k < r$. Similarly, if $\gamma = (i_1\ j_2\ \ldots\ j_s)$, then $j_{k+1} = \gamma^k(i_1)$ for $k < s$. Without loss of generality, assume $r \leq s$. We have $j_{r+1} = \gamma^r(i_1) = \beta^r(i_1) = i_{r+1} = i_1$. This implies that β and γ must have the same length, i.e., $r = s$, and that $j_k = i_k$ for all k, and so $\beta = (i_1\ i_2\ \ldots\ i_r) = \gamma$.

And the proofs are complete. ⌑

Lemma 1.24. *If $k, \ell \geq 0$ and the letters a, b, c_i, d_j are all distinct, then*

$$(a\ b)\,(a\ c_1\ \ldots\ c_k\ b\ d_1\ \ldots\ d_\ell) = (a\ c_1\ \ldots\ c_k)\,(b\ d_1\ \ldots\ d_\ell)$$

and

$$(a\ b)\,(a\ c_1\ \ldots\ c_k)\,(b\ d_1\ \ldots\ d_\ell) = (a\ c_1\ \ldots\ c_k\ b\ d_1\ \ldots\ d_\ell)\,.$$

Proof. The left side sends $a \mapsto c_1 \mapsto c_1$; $c_1 \mapsto c_{i+1} \mapsto c_{i+1}$ if $i < k$; $c_k \mapsto b \mapsto a$; $b \mapsto d_1 \mapsto d_1$; $d_j \mapsto d_{j+1} \mapsto d_{j+1}$ if $j < \ell$; $d_\ell \mapsto a \mapsto b$. Similar evaluation of the right side shows that both permutations are equal. For the second equation, just multiply both sides of the first equation by $(a\ b)$ on the left. ⌑

Lemma 1.25. *If $\alpha, \tau \in S_n$ and τ is a transposition, then* $\operatorname{sgn}(\tau\alpha) = -\operatorname{sgn}(\alpha)$.

Proof. Let $\alpha = \beta_1 \dots \beta_t$ be a complete factorization of α into disjoint cycles and $\tau = (a\ b)$. If a and b occur in the same cycle, since disjoint cycles commute, assume they occur in β_1 without loss of generality. Then, $\beta_1 = (a\ c_1\ \dots\ c_k\ b\ d_1\ \dots\ d_\ell)$. We have $\tau\beta_1 = (a\ b)\,(a\ c_1\ \dots\ c_k\ b\ d_1\ \dots\ d_\ell)$. By Lemma 1.24, $\tau\beta_1 = (a\ c_1\ \dots\ c_k)\,(b\ d_1\ \dots\ d_\ell)$, so the complete factorization $\tau\alpha = (\tau\beta_1)\dots\beta_t$ has 1 more disjoint cycle than α. Therefore, $\operatorname{sgn}(\tau\alpha) = (-1)^{n-(t+1)} = -\operatorname{sgn}(\alpha)$. If a and b occur in two different cycles, say β_1 and β_2, then $\beta_1 = (a\ c_1\ \dots\ c_k)$ and $\beta_2 = (b\ d_1\ \dots\ d_\ell)$. We have $\tau\beta_1\beta_2 = (a\ b)\,(a\ c_1\ \dots\ c_k)\,(b\ d_1\ \dots\ d_\ell)$. By Lemma 1.24, $\tau\beta_1\beta_2 = (a\ c_1\ \dots\ c_k\ b\ d_1\ \dots\ d_\ell)$, so the complete factorization $\tau\alpha = (\tau\beta_1\beta_2)\dots\beta_t$ has 1 fewer disjoint cycle than α. Therefore, $\operatorname{sgn}(\tau\alpha) = (-1)^{n-(t-1)} = -\operatorname{sgn}(\alpha)$. ⌑

Exercises

1.1. Let $\alpha, \beta \in S_n$ be disjoint.

(i) Prove, for all i, that α moves i if and only if α^{-1} moves i.
(ii) Prove, if $\alpha\beta = \mathbb{1}$, then $\alpha = \mathbb{1}$ and $\beta = \mathbb{1}$.
(iii) Prove, for all $k \geq 1$, $(\alpha\beta)^k = \alpha^k\beta^k$. Indeed, prove that if α and β are permutations that commute, then $(\alpha\beta)^k = \alpha^k\beta^k$ for all $k \geq 1$.

Proof. Here are the proofs:

(i) Suppose $\alpha(i) = j \neq i$. We have $\alpha^{-1}(j) = i$. Suppose, for contradiction, that α^{-1} fixes i. Then, $\alpha^{-1}(i) = \alpha^{-1}(j) = i$, contradicting the injectivity of α^{-1}. The same argument applies for the converse.
(ii) Let i be arbitrary. Since $\alpha\beta = \mathbb{1}$, we have $\alpha(\beta(i)) = i$ and $(\alpha\beta)^{-1}(i) = \beta^{-1}(\alpha^{-1}(i)) = i$. Suppose, for contradiction, that β moves i. Then, α fixes i, α^{-1} fixes i, and β^{-1} moves i by the previous part. We have $\beta^{-1}(\alpha^{-1}(i)) = \beta^{-1}(i) \neq i$, which is a contradiction. The same argument applies for α. Therefore, neither moves i, and since i was arbitrary, $\alpha = \beta = \mathbb{1}$.
(iii) Let $\alpha = \gamma_1 \dots \gamma_m$ and $\beta = \triangle_1 \dots \triangle_\ell$ be complete factorizations of α and β into disjoint cycles. Let i be arbitrary. If α moves i, let γ_1 be the cycle to move i. We know no other γ_i's moves i, and β does not move i. Then, $(\alpha\beta)^k(i) = (\gamma_1 \dots \gamma_m \triangle_1 \dots \triangle_\ell)^k(i) = \gamma_1^k(i)$. On the other hand, we have also $\alpha^k(i)\beta^k(i) = \gamma_1^k(i)$. The same argument applies if β moves i. If neither moves i, it is clear that $(\alpha\beta)^k(i) = \alpha^k(i)\beta^k(i) = i$.

To generalize the proof to commuting permutations, we use a proof by induction on k. For the base case $k = 1$, the claim is obviously true. Suppose for $k - 1 \geq 1$, $(\alpha\beta)^{k-1} = \alpha^{k-1}\beta^{k-1}$. We show $(\alpha\beta)^k = \alpha^k\beta^k$. We have $(\alpha\beta)^k = (\alpha\beta)^{k-1}(\alpha\beta)$. The inductive hypothesis yields $(\alpha\beta)^{k-1}(\alpha\beta) = \alpha^{k-1}\beta^{k-1}\alpha\beta$. Since α and β commute, their powers commute as well. This makes $\alpha^{k-1}\beta^{k-1}\alpha\beta = \alpha^k\beta^k$, which is what we want to show.

And we are done. ⌑

2
Groups

Definitions

Definition 2.1 (Operation)**.** An *operation* on a set G is a function $\circ : G \times G \to G$. Since writing $\circ(a_1, a_2)$ is a bit awkward, we write $a_1 \circ a_2$ and can abbreviate it to $a_1 a_2$.

Definition 2.2 (Groups)**.** A *group* is a set G with an operation $\circ : G \times G \to G$ that satisfies

(i) *Associative Law:* For all $x, y, z \in G$, $(x \circ y) \circ z = x \circ (y \circ z)$.
(ii) *Existence of Identity:* There exists (uniquely) $\mathbb{1} \in G$ such that for all $a \in G$, $a \circ \mathbb{1} = a = \mathbb{1} \circ a$.
(iii) *Existence of Inverse:* For all $a \in G$, there exists (uniquely) $a^{-1} \in G$ such that $a \circ a^{-1} = \mathbb{1} = a^{-1} \circ a$.

We denote this group as $(G, \circ)$, or abbreviated as G.

Definition 2.3 (Abelian Groups)**.** A group $(G, \circ)$ is called *abelian* if for all $x, y \in G$, $x \circ y = y \circ x$.

Definition 2.4 (Powers)**.** Let G be a set with operation $\circ$. Define the powers a^n inductively:

$$a^1 = a \quad \text{and} \quad a^n = a \circ a^{n-1}$$

Definition 2.5. An expression $a_1 a_2 \dots a_n$ *needs no parentheses* if, no matter what choices of the orders of multiplication of adjacent factors are made, all the resulting elements of G are equal.

Definition 2.6. If $(G, \circ)$ is a group and $a \in G$, define $a^0 = \mathbb{1}$. If $n > 0$, $a^{-n} = (a^{-1})^n$.

Definition 2.7 (Order)**.** Let $(G, \circ)$ be a group and $a \in G$. If $a^k = \mathbb{1}$ for some $k \in \mathbb{N}^*$, then the smallest such k is called the *order* of a. If no such k exists, a has *infinite order*.

Definition 2.8 (Subgroups)**.** Let $(G, \circ)$ be a group and $H \subset G$. H is called a *subgroup* if it satisfies:

(i) *Existence of Identity:* $\mathbb{1} \in H$
(ii) *Closure:* For all $x, y \in H$, $x \circ y \in H$.
(iii) *Existence of Inverse:* If $x \in H$, then $x^{-1} \in H$.

Definition 2.9 (Cyclic Subgroups)**.** If $(G, \circ)$ is a group and $a \in G$, write

$$\langle a \rangle = \{a^n : n \in \mathbb{Z}\} = \{\text{all powers of } a\};$$

$\langle a \rangle$ is called the *cyclic subgroup* of G generated by a.

Definition 2.10 (Group Order)**.** If G is a finite group, then the number of elements in G, denoted by $|G|$, is called the order of G.

Definition 2.11 (Coset)**.** Let G be a group and $a \in G$. If H is a subgroup of G, then the (left) coset aH is the subset $aH \subset G$, defined by $aH = \{ah : h \in H\}$.

Definition 2.12 (Subgroup Index)**.** The *index* of a subgroup H of a group G is the number of cosets of H in G, denoted by $[G : H]$. It follows from Theorem 2.18 that $[G : H] = |G|/|H|$.

Definition 2.13 (Cyclic Group)**.** A group G is cyclic if there is an element $a \in G$ such that $G = \langle a \rangle$. In this case, G consists of all the powers of a.

Theorems

Theorem 2.14 (Generalized Associativity)**.** *If G is a set with an associative operation $\circ$ and if $a_1, a_2, \ldots, a_n \in G$, then the expression $a_1 a_2 \ldots a_n$ needs no parentheses.*

Proof. We show this using complete induction. Since the result is vacuously true for $n = 1$ and $n = 2$, we need only start the base case from $n = 3$. At $n = 3$, $(a_1 a_2) a_3 = a_1 (a_2 a_3)$ by associativity of $\circ$.

Now, assume for all $3 \leq k \leq n$, the expressions $a_1 a_2 \ldots a_k$ needs no parentheses. We show $a_1 a_2 \ldots a_{n+1}$ needs no parentheses. Consider 2 elements obtained from an expression $a_1 a_2 \ldots a_{n+1}$ after two different series of choices

$$(a_1 a_2 \ldots a_i)(a_{i+1} \ldots a_{n+1}) \quad \text{and} \quad (a_1 a_2 \ldots a_j)(a_{j+1} \ldots a_{n+1}).$$

Without loss of generality, assume $i \leq j$. Since each of the 4 expressions has fewer terms than n, each needs no parentheses by the inductive hypothesis. So, if $i = j$, it follows that the two products give the same element of G. Hence, assume $i < j$. Then, the first expression can be rewritten as

$$(a_1 a_2 \ldots a_i)(a_{i+1} \ldots a_{n+1}) = (a_1 a_2 \ldots a_i)[(a_{i+1} \ldots a_j)(a_{j+1} \ldots a_n)]$$

and the second can be rewritten as

$$[(a_1 a_2 \ldots a_i)(a_{i+1} \ldots a_j)](a_{j+1} \ldots a_{n+1})$$

where each smaller expression also needs no parentheses. Thus, these expressions yield unique A, B, C respectively. The first expression yields $A(BC)$ and the second yields $(AB)C$. By associativity of $\circ$, these two expressions are equal, completing the proof. ⌑

Theorem 2.15 (Laws of Exponents)**.** *Let $(G, \circ)$ be a group $a, b \in G$, and $r, s \in \mathbb{Z}$. Then, the following statements hold:*

(i) $(ab)^{-1} = b^{-1}a^{-1}$.

(ii) *If* a, b *commute, then* $(ab)^r = a^r b^r$.

(iii) $(a^{-1})^r = (a^r)^{-1} = a^{-r}$.

(iv) $(a^r)^s = a^{rs}$.

(v) $a^r a^s = a^{r+s}$.

Proof. The proof is trivial but long and tedious, hence left to the reader. ⌑

Theorem 2.16. *Let* $(G, \circ)$ *be an abelian group* $a, b \in G$*, and* $r, s \in \mathbb{Z}$*. Then, the following statements hold:*

(i) $r(a \circ b) = ra \circ rb$.

(ii) $r(sa) = (rs)a$.

(iii) $ra \circ sa = (r \circ s)a$.

Proof. The proof is trivial but long and tedious, hence left to the reader. ⌑

Theorem 2.17. *Let* G *be a finite group and let* $a \in G$*. Then the order of* a *is the number of elements in* $\langle a \rangle$*.*

Proof. Since G is finite, there exists $k \geq 1$ such that $\mathbb{1}, a, a^2, \dots, a^{k-1}$ consists of k distinct elements, while $\mathbb{1}, a, a^2, \dots, a^{k-1}, a^k$ has a repetition. Therefore, $a^k \in \{\mathbb{1}, a, a^2, \dots, a^{k-1}\}$, which means $a^k = a^i$ for some $0 \leq i < k$, and $a^{k-i} = \mathbb{1}$. If $i > 0$, that means there exists $j = k - i$ such that $a^j = \mathbb{1}$ and $j \neq 0$, which contradicts the fact that the first list has no repetitions. Therefore, $i = 0$, and k is the order of a.

If $H = \{\mathbb{1}, a, a^2, \dots, a^{k-1}\}$, then $|H| = k$, which is the order of a. Therefore, it suffices to show that $H = \langle a \rangle$. By definition of H, $H \subset \langle a \rangle$. For the converse, let $a^i \in \langle a \rangle$ be arbitrary. By division algorithm, $i = qk + r$ for some $0 \leq r < k$. Hence, $a^i = (a^k)^q a^r = a^r \in H$. Therefore, $\langle a \rangle \subset H$, which means $\langle a \rangle = H$. ⌑

Theorem 2.18 (Lagrange's Theorem)**.** *Let* G *be a finite group. If* H *is a subgroup of* G*, then* $|H|$ *is a divisor of* $|G|$*.*

Proof. Let $\{a_i H\}_{i=1}^k$ be the set of all distinct cosets of H in G. Then

$$G = a_0 H \cup a_1 H \cup \cdots \cup a_k H$$

since each $g \in G$ lies in gH, and $gH = a_i H$ for some i. Since each sets are distinct, by Lemma 2.21, they are disjoint. Also by Lemma 2.21, it follows that

$$|G| = \sum_{i=0}^{k} |a_i H| = \sum_{i=0}^{k} |H| = k|H|,$$

which is what we want to show. It follows from this theorem that $|G| = [G : H]|H|$. ⌑

Lemmas

Lemma 2.19. *Let $(G, \circ)$ be a group. The following laws hold:*

(i) Cancellation Law: If $x \circ a = x \circ b$ or $a \circ x = b \circ x$, then $a = b$.
(ii) Uniqueness of Identity: The existence of $\mathbb{1} \in G$ is unique.
(iii) Uniqueness of Inverse: For all $x \in G$, there exists a unique x^{-1} such that $x \circ x^{-1} = \mathbb{1} = x^{-1} \circ x$.
(iv) Cancellation of Inverses: $(x^{-1})^{-1} = x$ for all $x \in G$.

Proof. The proof is trivial and left to the reader. ⌑

Lemma 2.20. *Let $(G, \circ)$ be a group and assume that $a \in G$ has finite order k. If $a^n = \mathbb{1}$, then $k \mid n$. In fact, $\{n \in \mathbb{Z} : a^n = \mathbb{1}\}$ is the set of all the multiples of k.*

Proof. We first show that if $a^n = \mathbb{1}$, then $k \mid n$. By the Division Algorithm, there exist unique integers q and r such that $n = qk + r$ where $0 \leq r < k$. Using the laws of exponents and the fact that $a^k = \mathbb{1}$, we can expand a^n as:

$$a^n = a^{qk+r} = (a^k)^q \circ a^r = (\mathbb{1})^q \circ a^r = a^r.$$

Since we are given that $a^n = \mathbb{1}$, it follows that $a^r = \mathbb{1}$.

If $r > 0$, the fact that $r < k$ and $a^r = \mathbb{1}$ contradicts the minimality of k. Therefore, $r = 0$. Substituting $r = 0$ back into our division equation yields $n = qk$, which means $k \mid n$. Thus, every element in the set $\{n \in \mathbb{Z} : a^n = \mathbb{1}\}$ is a multiple of k. Conversely, suppose $n = mk$, for some $m \in \mathbb{Z}$. Then $a^n = a^{mk} = (a^k)^m = \mathbb{1}^m = \mathbb{1}$. This implies that $n \in \{n \in \mathbb{Z} : a^n = \mathbb{1}\}$, completing the proof. ⌑

Lemma 2.21. *Let G be a group, $a, b \in G$, and H be a subgroup of G. Then the following hold:*

(i) $aH = bH$ if and only if $b^{-1}a \in H$. In particular, $aH = H$ if and only if $a \in H$.
(ii) If $aH \cap bH \neq \emptyset$, then $aH = bH$.
(iii) $|aH| = |H|$ for all $a \in G$.

Proof. Here are some proof sketches.

(i) First, we assume $aH = bH$. Let $ah_0 \in aH$ be arbitrary. Then, $ah_0 \in bH$, and $ah_0 = bh$ for some $h \in H$. Therefore, $b^{-1}a = hh_0^{-1} \in H$. Conversely, assume $b^{-1}a = h_0 \in H$. Then, $a = bh_0$ and $b = ah_0^{-1}$. If $x \in aH$, $x = ah = bh_0h$ for some $h \in H$, so $x \in bH$. If $x \in bH$, $x = bh = ah_0^{-1}h$ for some $h \in H$, so $x \in aH$. Therefore, $aH = bH$. The final conclusion follows trivially since $aH = \mathbb{1}H$.
(ii) Let $x \in aH \cap bH$. Then, $x = ah_0 = bh_1$, which means $b^{-1}a = h_1h_0^{-1} \in H$. Therefore, $aH = bH$.
(iii) Define $f : H \to aH$ as $f(h) = ah$. We can easily find its inverse $g : aH \to H$ as $g(ah) = a^{-1}h$. This is bijective so $|H| = |aH|$.

And the proofs are complete. ⌑

Lemma 2.22 (Order of Product of Commuting Elements)**.** *Let G be a finite abelian group and $a, b \in G$ with orders m, n respectively. If* $\gcd(m, n) = 1$*, then* $\mathrm{ord}(ab) = mn$*.*

Proof. Since $\gcd(m, n) = 1$, $|\langle a\rangle \cap \langle b\rangle| = 1$ since it is a divisor of $\gcd(m, n)$. Let $k = \mathrm{ord}(ab)$. Since G is abelian, $(ab)^{mn} = a^{mn}b^{mn} = (a^m)^n(b^n)^m = \mathbb{1}^n \cdot \mathbb{1}^m = \mathbb{1}$. This implies $k \mid mn$. Conversely, because $(ab)^k = \mathbb{1}$ and G is abelian, we have $a^k b^k = \mathbb{1}$, which implies $a^k = b^{-k}$. Since $a^k \in \langle a\rangle$ and $b^{-k} \in \langle b\rangle$, this element lies in their intersection. Thus, $a^k = b^{-k} = \mathbb{1}$. It follows immediately that $m \mid k$ and $n \mid k$, so $mn \mid k$ since $\gcd(m, n) = 1$. Since k and mn are positive integers that divide each other, we conclude $k = mn$. ⌑

Lemma 2.23. *A group S_n has order $n!$.*

Proof. We show this using induction. When $n = 1$, S_1 has only 1 element, which is the identity. Suppose S_k has $k!$ elements for some $k \geq 1$, we show that S_{k+1} has $(k + 1)!$ elements. Let $\alpha \in S_{k+1}$ be an arbitrary permutation. Consider the image of the element $k + 1$ under α, denoted $\alpha(k + 1)$, we have $\alpha(k + 1) = i$ where $i \in \{1, 2, \ldots, k, k + 1\}$. The restriction $\alpha|_{\{1,2,\ldots,k\}}$ must map bijectively onto the remaining k elements $\{1, 2, \ldots, k, k + 1\} \setminus \{i\}$. The number of ways to arrange these remaining k elements is precisely the number of permutations in S_k, which is $k!$. Since there are $k + 1$ mutually exclusive choices for the destination of $k + 1$, and for each choice there are $k!$ ways to permute the rest of the elements, $|S_{k+1}| = (k + 1)|S_k| = (k + 1)k! = (k + 1)!$. By the principle of mathematical induction, $|S_n| = n!$ for all $n \geq 1$. ⌑

Corollaries

Corollary 2.24. *Suppose G is a set with an associative operation $\circ$. If $a \in G$ and $m, n \geq 0$, then*

$$a^{m+n} = a^m a^n \quad \textit{and} \quad (a^m)^n = a^{mn}.$$

Proof. In the first expression, both sides arise from the expression having $m + n$ factors of a. In the second expression, both sides arise from the expression having mn factors of a. ⌑

Remark 2.25. It follows easily from Corollary 2.24 that

$$a^m a^n = a^{m+n} = a^{n+m} = a^n a^m \quad \text{and} \quad (a^m)^n = a^{mn} = a^{nm} = (a^n)^m.$$

In a group, negative powers can be defined.

Corollary 2.26. *If G is a finite group and $a \in G$, then the order of a is a divisor of $|G|$.*

Proof. Let k be the order of a. We have $k = |\langle a\rangle| = |H|$ where $H = \{\mathbb{1}, a, a^2 \ldots, a^{k-1}\}$ is a subgroup of G. Since $|H|$ is a divisor of $|G|$, so is k. ⌑

Corollary 2.27. *If p is prime, then every group G of order p is cyclic.*

Proof. Let G be a group and $|G| = p$. Let $a \neq \mathbb{1} \in G$, and let $H = \langle a \rangle$ be the cyclic subgroup generated by a. Since $|H|$ is a divisor of $|G| = p$ and $|H| > 1$, it follows that $|H| = p = |G|$. So, $H = G$ as desired. ⌑

Corollary 2.28. *Every cyclic group is abelian.*

Proof. Let $G = \langle a \rangle$ be a cyclic group. Let $h, k \in G$ be arbitrary. Then, $h = a^m$ and $k = a^n$ for some $m, n \in \mathbb{Z}$. We have $hk = a^m a^n = a^{m+n} = a^{n+m} = a^n a^m = kh$. Therefore, G is abelian. ⌑

Exercises

2.1. Prove that if H, K are subgroups of a group G and $\gcd(|H|, |K|) = 1$, then $H \cap K = \{\mathbb{1}\}$.

Proof. We show the contrapositive. Suppose $H \cap K \neq \{\mathbb{1}\}$. It is obvious that $|H \cap K| > 1$. Since $H \cap K$ is a subgroup of H, $|H \cap K|$ is a divisor of $|H|$. Similarly, $|H \cap K|$ is a divisor of $|K|$. Therefore, $|H \cap K|$ is a common divisor of $|H|$ and $|K|$. Therefore, $\gcd(|H|, |K|) \neq 1$. ⌑

2.2. Let $G = \langle a \rangle$ be a cyclic group of order n. Prove that a^k is a generator of G, i.e., $G = \left\langle a^k \right\rangle$ if and only if $\gcd(k, n) = 1$.

Proof. $\boxed{\Longrightarrow}$ Suppose $G = \left\langle a^k \right\rangle$. Since $a \in \langle a \rangle$, $a \in \left\langle a^k \right\rangle$, and $a = a^{jk}$ for some j. Therefore, $a^{jk-1} = \mathbb{1}$, and since G has order n, $n \mid (jk - 1)$. Therefore, $jk - 1 = mn$, and $jk - mn = 1$. Since there exists a linear combination of n and k such that it is 1, $\gcd(k, n) = 1$.

$\boxed{\Longleftarrow}$ Suppose $\gcd(k, n) = 1$. It follows easily that $\left\langle a^k \right\rangle \subset G$ since $a^{kj} \in \langle a \rangle$ for all j. Therefore, we need only show $G \subset \left\langle a^k \right\rangle$. If $a = a^{jk}$, it is obvious that $a^i = a^{ijk} \in \left\langle a^k \right\rangle$ for all i. Therefore, we need only show that $a \in \left\langle a^k \right\rangle$. Since $\gcd(k, n) = 1$, there exists $r, s \in \mathbb{Z}$ such that $rk + sn = 1$, which means $a^{rk+sn} = a$. It follows that $a^{rk} a^{sn} = a$. Since $|G| = n$, $a^n = \mathbb{1}$, and $a^{rk} a^{sn} = a^{rk}$. Therefore, $a = a^{rk}$, so $a \in \left\langle a^k \right\rangle$ as desired. ⌑

2.3. Prove that every subgroup S of a cyclic group $G = \langle a \rangle$ is itself cyclic.

Proof. If $S = \{\mathbb{1}\}$, the claim is trivially true. If S is not trivial, let $k = \min\{k \in \mathbb{N}^* : a^k \in S\}$. We show that $S = \left\langle a^k \right\rangle$. To see that $\left\langle a^k \right\rangle \subset S$, observe that every $a^{jk} \in S$ by closure of S. Conversely, suppose, for contradiction, there exists $i \in \mathbb{Z}$ such that $a^i \in S$ but $a^i \notin \left\langle a^k \right\rangle$. By division algorithm, we have $a^i = a^{qk} a^r$ for some $0 < r < k$. But then, $a^r = a^i a^{-qk} \in S$, contradicting the minimality of k. Therefore, no such i exists, and the proof is complete. ⌑

2.4. Let G be a group of order 4. Prove that G is either cyclic or $x^2 = \mathbb{1}$ for all $x \in G$. Conclude that G must be abelian.

Proof. If there exists $a \in G$ such that $\text{ord}(a) = 4$, then G is cyclic. It follows from Corollary 2.28 that G is abelian. Otherwise, the order of any non-identity element must divide $|G| = 4$, meaning they must have order 2. Since $\mathbb{1}^2 = \mathbb{1}$, it follows that $x^2 = \mathbb{1}$ for all $x \in G$. This implies $x = x^{-1}$ for all $x \in G$. For any $a, b \in G$, we have $ab = (ab)^{-1} = b^{-1}a^{-1} = ba$, and G is abelian. ⌑

2.5. Prove that every abelian group G of order 6 is cyclic.

Proof. We show that there exists an element with order 6. First, we need to show that there exists at most 1 element with order 2. Therefore, suppose for contradiction, that there exists $a, b \in G$ such that $a \neq b$ but $\text{ord}(a) = \text{ord}(b) = 2$. We know that $ab \neq a$ and $ab \neq b$, but $(ab)^2 = a^2b^2 = \mathbb{1}$. Therefore, $H = \{1, a, b, ab\}$ is a subgroup of G. However, the order of H is 4, and $|H| \nmid |G|$. Contradiction. If $a \in G$ has order 3, then $a^3 = \mathbb{1}$, and $a^{-1} = a^2$. It is easy to check that $\text{ord}(a^{-1}) = 3$ as well. Therefore, elements of order 3 come in pairs. If there are 0 or 2 elements with order 3 and no elements of order 6, $|G| < 6$, contradiction. Therefore, either there exists some elements with order 6, or there are 4 elements of order 3 and 1 of order 2. G is obviously cyclic in the first case, so consider the second case. Pick $x, y \in G$ such that $\text{ord}(x) = 2$ and $\text{ord}(y) = 3$. By Lemma 2.22, $\text{ord}(xy) = 6$, and G is cyclic. ⌑

3

Geometry

Definitions

Definition 3.1 (Motion)**.** A function $\varphi : \mathbb{R}^2 \to \mathbb{R}^2$ is called a *motion* if it is *distance preserving*, i.e., $\|\varphi(P) - \varphi(Q)\| = \|P - Q\|$ for all points $P, Q \in \mathbb{R}^2$.

Definition 3.2 (Symmetry Group)**.** The *symmetry group* $\Sigma(\Omega)$ of a figure Ω in the plane is the set of all motions φ with $\varphi(\Omega) = \Omega$. The elements of $\Sigma(\Omega)$ are called *symmetries* of Ω.

Remark 3.3. It is easy to see that $\Sigma(\Omega)$ is a subgroup of $\mathfrak{M}$, where $\mathfrak{M}$ is the set of all motions of the plane, which means it is a group.

Definition 3.4 (Dihedral Group)**.** Let π_n be a regular polygon with n vertices and centered at $\vec{0}$. The symmetry group $\Sigma(\pi_n)$ is called the *dihedral group* of order $2n$ and is denoted by D_{2n}.

Definition 3.5 (Affine Map)**.** A function $\alpha : \mathbb{R}^2 \to \mathbb{R}^2$ is an *affine map* if there exists a non-singular linear transformation T and a point $V \in \mathbb{R}^2$ such that $\alpha(P) = T(P) + V$ for every $P \in \mathbb{R}^2$. The set of all affine maps is denoted by $\mathrm{Aff}(2, \mathbb{R})$.

Definition 3.6 (Convex Combination)**.** A *convex combination* of $P, Q \in \mathbb{R}^2$ is an $\mathbb{R}-$linear combination of the form $U_t = tP + (1-t)Q$, where $0 \le t \le 1$. We call $U_t = tP + (1-t)Q$ the *t-point* of $\overline{PQ}$.

Definition 3.7 (Affine Invariant)**.** A geometric property is called an *affine invariant* if it is preserved by every affine map φ. A *euclidean invariant* is a property that is preserved by every motion.

Remark 3.8. For every subgroup G is the symmetric group $S_{\mathbb{R}\times\mathbb{R}}$, F. Klein, in his Erlangen Program in 1972, defined its *geometry* to be the study of all those properties that are left invariant by every $\varphi \in G$.

Remark 3.9. If G is the group of all *homeomorphisms*, the corresponding geometry is called *topology*. Some topological invariants are connectedness and compactness. To learn about this topic, I recommend *Topology (MA331)*, a wonderful Colby course that I was fortunate enough to take.

Theorems

Theorem 3.10. *The following statements hold true:*

(i) Every motion φ is a bijection.

(ii) The set $\mathfrak{M}$ of all motions of the plane is a group under composition.

Proof. Here are the proofs:

(i) If $P \neq Q$, $\|P - Q\| \neq 0$, which forces $\|\varphi(P) - \varphi(Q)\| \neq 0$ since φ is distance preserving. This implies $\varphi(P) \neq \varphi(Q)$, and φ is injective.
Now, let $V \in \mathbb{R}^2$. If ℓ is the $x-$axis and ℓ' is the $y-$axis, then $\varphi(\ell)$ and $\varphi(\ell')$ are perpendicular, by Lemma 3.16. If $V \in \varphi(\ell) \cup \varphi(\ell')$, then $V \in \operatorname{im}\varphi$. If $V \notin \varphi(\ell) \cup \varphi(\ell')$, choose a line λ' through V that intersects both $\varphi(\ell)$ and $\varphi(\ell')$, say, $\lambda' \cap \varphi(\ell) = \{\varphi(P)\}$ and $\lambda' \cap \varphi(\ell') = \{\varphi(Q)\}$. If λ is the line through P and Q, then $\varphi(\lambda)$ is determined by the line passing through $\varphi(P)$ and $\varphi(Q)$ by Lemma 3.16, that is, $\varphi(\lambda) = \lambda'$. Therefore, $V \in \lambda' = \varphi(\lambda) \subset \operatorname{im}\varphi$, and φ is surjective.

(ii) We show that $\mathfrak{M}$ is a subgroup of $S_{\mathbb{R}^2}$, the subgroup of all permutations of $\mathbb{R}^2$. First, the identity function $\mathbb{1}$ is clearly a motion. If φ and φ' are motions, then, for all P, Q, we have

$$\|\varphi'(\varphi(P)) - \varphi'(\varphi(Q))\| = \|\varphi(P) - \varphi(Q)\| = \|P - Q\|.$$

Therefore, compositions of motions are also motions. Finally, since each motion φ is bijective, there exists an inverse φ^{-1}, which can be defined as a motion

$$\|P - Q\| = \|\varphi(\varphi^{-1}(P)) - \varphi(\varphi^{-1}(Q))\| = \|\varphi^{-1}(P) - \varphi^{-1}(Q)\|.$$

And we are done. ⌑

Theorem 3.11. $\operatorname{Aff}(2, \mathbb{R})$ *is a group under composition.*

Proof. We show that $\operatorname{Aff}(2, \mathbb{R})$ is a subgroup of $S_{\mathbb{R}^2}$, the group of all permutations of $\mathbb{R}^2$. We can see that $\mathbb{1}$ is obviously an affine map. If α and α' are affine, we have

$$\alpha'\alpha(P) = \alpha'(T(P) + V) = T'(T(P) + V) + V' = T'T(P) + T'T(V) + V',$$

and so $\alpha'\alpha$ is an affine map with linear transformation $T'T$ and point $T'(V) + V'$. Finally, the inverse of α is given by $\alpha^{-1}(P) = T^{-1}(P) - T^{-1}(V)$. ⌑

Remark 3.12. We can easily see that $\operatorname{Aff}(n, \mathbb{R})$ is also a group.

Theorem 3.13. *If $\triangle$ and $\triangle'$ are triangles, then there exists an affine map $\alpha : \mathbb{R}^2 \to \mathbb{R}^2$ such that $\alpha(\triangle) = \triangle'$.*

Proof. Let the vertices of $\triangle$ be P, Q, U. We know there is a linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ with $T(\varepsilon_1) = P - U$ and $T(\varepsilon_2) = Q - U$. If $\alpha : \mathbb{R}^2 \to \mathbb{R}^2$ is the affine map defined by $\alpha(V) = T(V) + U$, then $\alpha(\vec{0}) = U$, $\alpha(\varepsilon_1) = P$, and $\alpha(\varepsilon_2) = Q$, that is, α carries the triangle Φ with vertices $\vec{0}, \varepsilon_1, \varepsilon_2$ onto the triangle $\triangle$. Similarly, if the vertices of $\triangle'$ are P', Q', U', then there exists an affine map α' carrying the triangle Φ onto the triangle $\triangle'$. Therefore, $\alpha'\alpha^{-1}$ is the affine map carrying $\triangle$ onto $\triangle'$. ⌑

Theorem 3.14. *For every triangle $\triangle$, the medians meet in a common point that is the $\frac{2}{3}-$point on each of the median.*

Proof. We first show that the theorem is true in the special case of an equilateral triangle π_3. Hence, let $\triangle PQU$ be such a triangle, and let L, M be the $\frac{1}{2}$−points of $\overline{PU}$ and $\overline{QU}$, respectively. Let $\overline{PM} \cap \overline{QL} = O$. The medians $\overline{PM}, \overline{QL}$ are angle bisectors, so $\triangle OPQ$ is isosceles with base angles 30°. Therefore, $\overline{OP}$ and $\overline{OQ}$ have the same length, say, b. It follows that $\overline{OL}$ and $\overline{OM}$ have the same length, say, h. The Pythagorean theorem applied to $\triangle PMQ$ gives $(b+h)^2 + a^2 = 4a^2$, hence $(b+h)^2 = 3a^2$, while the Pythagorean theorem applied to $\triangle OMQ$ gives $h^2 + a^2 = b^2$, hence $b^2 - h^2 = a^2$. Thus, $(b+h)^2 = 3(b+h)(b-h)$, so that $b + h = 3(b-h)$, and $b = 2h$. Therefore, two medians of $\triangle PQU$ meet at the $\frac{2}{3}$−point of either of them, for the medians have the same length. It follows that the three medians meet at this one point. By Theorem 3.13, there exists an affine map α with $\alpha(\pi_3) = \triangle$. By Lemma 3.18, α preserves line segments, medians, and $\frac{2}{3}$−points, and so the property stated in the theorem is inherited by $\triangle$ from π_3. ⌑

Theorem 3.15. *If E is the ellipse with equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ where $a, b > 0$, then the area of the region bounded by E, call it $\tilde{E}$, is πab.*

Proof. Let D be the unit circle whose equation is $x^2 + y^2 = 1$. We know from standard geometry that the circular region enclosed by D, call it $\tilde{D}$, has an area of π. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a transformation defined by $T(x, y) = (ax, by)$. This transformation is nonsingular because $a, b \neq 0$, meaning its inverse is simply $T^{-1}(u, v) = \left(\frac{1}{a}u, \frac{1}{b}v\right)$. If a point (x, y) lies on the boundary circle D, its coordinates satisfy $x^2 + y^2 = 1$. Substituting the transformed coordinates $(u, v) = (ax, by)$, we get:

$$1 = x^2 + y^2 = \frac{u^2}{a^2} + \frac{v^2}{b^2}$$

This is exactly the equation of the ellipse E. Thus, T maps the boundary circle D onto the ellipse E, and consequently, it maps the entire enclosed region $\tilde{D}$ onto the elliptical region $\tilde{E}$.

Next, we consider how T changes areas. Let R be a rectangle with height h, width w, and sides parallel to the x- and y-axes. The transformation T scales the x-coordinates by a and the y-coordinates by b. Because it scales the axes independently, it preserves right angles and parallel lines. Therefore, $T(R)$ is a new rectangle with width aw and height bh. The area of this new rectangle is

$$\text{area}(T(R)) = (aw)(bh) = ab(wh) = ab\,\text{area}(R).$$

To calculate the exact area of the circular region $\tilde{D}$, we can approximate it using a grid of tiny rectangles $\mathscr{G}$, where each rectangle has width Δx and height Δy. Let A_k be the sum of the areas of all rectangles in $\mathscr{G}$ that lie completely inside $\tilde{D}$:

$$A_k = \sum_{R_i \subset \tilde{D}} \text{area}(R_i)$$

As the grid spacing Δx and Δy approaches zero, these approximations converge to the exact area of the circle, meaning $\lim A_k = \text{area}(\tilde{D}) = \pi$. Because T is a continuous scaling transformation, applying T

to our grid $\mathscr{G}$ creates a new rectangular grid $T(\mathscr{G})$ that perfectly approximates the elliptical region $\tilde{E}$. Every original rectangle R_i inside the circle corresponds to a scaled rectangle $T(R_i)$ inside the ellipse. Since each new rectangle has an area exactly ab times larger than the original, the total approximated area for the ellipse is:

$$\sum_{T(R_i)\subset\tilde{E}} \operatorname{area}(T(R_i)) = \sum_{R_i\subset\tilde{D}} ab\,\operatorname{area}(R_i) = ab \sum_{R_i\subset\tilde{D}} \operatorname{area}(R_i) = abA_k.$$

Taking the limit as the grid spacing approaches zero, the scaling constant ab factors cleanly out of the limit:

$$\operatorname{area}(\tilde{E}) = \lim_{\Delta x,\Delta y\to 0}(abA_k) = ab\left(\lim_{\Delta x,\Delta y\to 0} A_k\right) = ab\,\operatorname{area}(\tilde{D}).$$

Since $\operatorname{area}(\tilde{D}) = \pi$, we get $\operatorname{area}(\tilde{E}) = \pi ab$, and the proof is complete. ⌑

Lemmas

Lemma 3.16. *Let φ be a motion. The following statements hold:*

(i) If $\overline{PQ}$ is the line segment with endpoints P and Q, then $\varphi(\overline{PQ}) = \overline{\varphi(P)\varphi(Q)}$. If ℓ is the line containing P and Q, then $\varphi(\ell)$ is the line containing $\varphi(P)$ and $\varphi(Q)$.

(ii) If $\triangle$ is the triangle with vertices P, Q, U, then $\varphi(\triangle)$ is the triangle with vertices $\varphi(P), \varphi(Q), \varphi(U)$.

(iii) If ℓ and ℓ' are perpendicular lines, then $\varphi(\ell)$ and $\varphi(\ell')$ are also perpendicular.

Proof. Here are the proofs:

(i) $\boxed{\subset}$ First, we show that for all $U \in \overline{PQ}$, $\varphi(U) \in \overline{\varphi(P)\varphi(Q)}$. Therefore, let $U \in \overline{PQ}$ be arbitrary and let $d = \|P - Q\|$. We have $\|P - U\| = t > 0$ and $\|Q - U\| = d - t$. If $\varphi(U) \notin \overline{\varphi(P)\varphi(Q)}$, the triangle inequality yields a contradiction:

$$d = \|\varphi(P) - \varphi(Q)\| < \|\varphi(P) - \varphi(U)\| + \|\varphi(U) - \varphi(Q)\| = t + (d - t) = d$$

$\boxed{\supset}$ Conversely, we show that for all $V \in \overline{\varphi(P)\varphi(Q)}$, there exists a $U \in \overline{PQ}$ such that $V = \varphi(U)$. If $\|\varphi(P) - V\| = s$, let $U \in \overline{PQ}$ such that $\|P - U\| = s$. Then $\varphi(U) = V$, since there exists a unique point on $\overline{\varphi(P)\varphi(Q)}$ whose distance to $\varphi(P)$ is s.

We can now show that if ℓ passes through P and Q, then $L = \varphi(\ell)$ passes through $\varphi(P)$ and $\varphi(Q)$.

$\boxed{\subset}$ Let ℓ be the line that passes through P and Q, and let $U \in \ell$. Then one of the points P, Q, U lies between the other two. Without loss of generality, assume Q lies between P and U. Then, $\|P - U\| = \|P - Q\| + \|Q - U\|$. Since φ preserves distances, $\|\varphi(P) - \varphi(U)\| = \|\varphi(P) - \varphi(Q)\| + \|\varphi(Q) - \varphi(U)\|$. Hence $\varphi(U)$ lies on the line through $\varphi(P)$ and $\varphi(Q)$. Since U was arbitrary, $\varphi(\ell) \subset L$.

$\boxed{\supset}$ Conversely, let $V \in L$ be an arbitrary point. Then one of the points $\varphi(P), \varphi(Q), V$ lies between the other two. Without loss of generality, assume $\varphi(Q)$ lies between $\varphi(P)$ and V. Then, $\|\varphi(P) - V\| = \|\varphi(P) - \varphi(Q)\| + \|\varphi(Q) - V\|$. Let $s = \|\varphi(Q) - V\|$. Let $U \in \ell$ such that Q lies between P and U, and $\|Q - U\| = s$. Because Q is between P and U, $\|P-U\| = \|P-Q\| + \|Q-U\| = \|P-Q\| + s$. Since φ preserves distances, $\|\varphi(P) - \varphi(U)\| = \|P-U\| = \|P-Q\|+s = \|\varphi(P)-\varphi(Q)\|+s$. On the other hand, $\|\varphi(Q)-\varphi(U)\| = \|Q-U\| = s$, which implies that $\varphi(Q)$ lies between $\varphi(P)$ and $\varphi(U)$ on the line L. Since $V \in L$ and the distance from $\varphi(Q)$ to both V and $\varphi(U)$ is s, uniqueness of points on a line dictates that $\varphi(U) = V$. Since $U \in \ell$, it follows that $V \in \varphi(\ell)$, and therefore $L \subset \varphi(\ell)$.

(ii) It follows from the previous part that each side of $\triangle$ corresponds to an edge in $\varphi(\triangle)$.

(iii) Let ℓ and ℓ' intersect in the point U, and choose points $P, Q \in \ell'$ equidistant from U. The line ℓ is characterized as being the set of all points equidistant from $\varphi(P)$ and $\varphi(Q)$, that is, $\varphi(\ell)$ is the perpendicular bisector of $\overline{\varphi(P)\varphi(Q)}$. But $\varphi(P)$ and $\varphi(Q)$ lie on the line $\varphi(\ell')$, so that $\varphi(\ell)$ is perpendicular to $\varphi(\ell')$.

And we are done. ⌑

Lemma 3.17. *If $P, Q \in \mathbb{R}^2$, then the line segment $\overline{PQ}$ consists of all vectors $U_t = tP + (1-t)Q$, where $0 \le t \le 1$.*

Proof. The family of all points of the form $U_t = t(Q-P) + P$ for all $t \in \mathbb{R}$ is the line containing $U_0 = P$ and $U_1 = Q$. If $0 < t < 1$, then U_t is the unique point on this line whose distance from P is $t\|Q-P\|$ and whose distance from Q is $(1-t)\|Q-P\|$. It follows that U_t is on the line segment $\overline{PQ}$. ⌑

Lemma 3.18. *Let α be an affine map. The following statements are true:*

(i) α preserves all convex combinations.

(ii) α preserves line segments: $\alpha(\overline{PQ}) = \overline{\alpha(P)\alpha(Q)}$ for all P, Q.

(iii) If U_t is the $t-$point of $\overline{PQ}$, where $0 \le t \le 1$, then $\alpha(U_t)$ is the $t-$point of $\overline{\alpha(P)\alpha(Q)}$. In particular, α preserves midpoints of line segments.

(iv) α preserves triangles: if $\triangle$ is the triangle with vertices P, Q, U, then $\alpha(\triangle)$ is a triangle with vertices $\alpha(P), \alpha(Q), \alpha(U)$.

(v) α preserves parallelism of lines.

Proof. Here are some proof sketches:

(i) Let $\alpha(P) = T(P) + V$ and let $U_t = tP + (1-t)Q$ be an arbitrary convex combination. We have

$$\begin{aligned}\alpha(U_t) &= T(tP + (1-t)Q) + V \\ &= tT(P) + (1-t)T(Q) + tV + (1-t)V \\ &= t(T(P) + V) + (1-t)(T(Q) + V) \\ &= t\alpha(P) + (1-t)\alpha(Q).\end{aligned}$$

(ii) Since $\overline{PQ}$ is the set of all convex combinations of P and Q, this follows immediately from (i).

(iii) This is proven by the equations in (i).

(iv) This follows from (ii). Each side of $\triangle$ corresponds to a side in $\alpha(\triangle)$.

(v) If ℓ and ℓ' are lines, so are $\alpha(\ell)$ and $\alpha(\ell')$. If $U \in \alpha(\ell) \cap \alpha(\ell')$, then $\alpha^{-1}(U) \in \ell \cap \ell'$, contradicting their being parallel.

And we are done. ⌑

Exercises

3.1. Prove that is φ is a motion and $\triangle$ is a triangle, then $\varphi(\triangle)$ and $\triangle$ are congruent. Conclude that motions also preserve angles, i.e., $\measuredangle ABC = \measuredangle\varphi(A)\varphi(B)\varphi(C)$.

Proof. Let the vertices of $\triangle$ be A, B, and C, so $\triangle = \triangle ABC$. By the definition of a motion, distance is preserved. Therefore, the side lengths of the transformed triangle $\varphi(\triangle)$ satisfy

$$\varphi(A)\varphi(B) = AB, \quad \varphi(B)\varphi(C) = BC, \quad \text{and} \quad \varphi(A)\varphi(C) = AC$$

By the SSS Congruence Postulate, $\triangle ABC \cong \triangle\varphi(A)\varphi(B)\varphi(C)$. Because corresponding parts of congruent triangles are congruent, it immediately follows that the corresponding angles are equal, so $\measuredangle ABC = \measuredangle\varphi(A)\varphi(B)\varphi(C)$, and the proof is complete. ⌑

3.2. If T is a nonsingular tranformation having matrix $\mu(T)$, and if Ω is a figure in the plane, prove that $\text{area}(T(\Omega)) = |\det(\mu(T))|\,\text{area}(\Omega)$.

Proof. Let the linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ be represented by the matrix $\mu(T) = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Let R be a simple rectangle in the plane with one vertex at the origin, spanned by the orthogonal vectors $\vec{u} = \begin{pmatrix} w \\ 0 \end{pmatrix}$ and $\vec{v} = \begin{pmatrix} 0 \\ h \end{pmatrix}$, where w is the width and h is the height. The area of this baseline rectangle is clearly $\text{area}(R) = wh$. The image of the rectangle $T(R)$ becomes a parallelogram spanned by the transformed vectors $T(\vec{u})$ and $T(\vec{v})$, where

$$T(\vec{u}) = \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} w \\ 0 \end{pmatrix} = \begin{pmatrix} aw \\ cw \end{pmatrix} \quad \text{and} \quad T(\vec{v}) = \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 0 \\ h \end{pmatrix} = \begin{pmatrix} bh \\ dh \end{pmatrix}.$$

The standard geometric formula for the area of a parallelogram spanned by two vectors $\vec{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ and $\vec{y} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$ is given by $\operatorname{area}(T(R)) = \left|\det \begin{pmatrix} aw & bh \\ cw & dh \end{pmatrix}\right|$. By the multicollinearity of $\mu(T)$, we have $\operatorname{area}(T(R)) = \left|wh \det \begin{pmatrix} a & b \\ c & d \end{pmatrix}\right| = wh|\det(\mu(T))| = |\det(\mu(T))| \operatorname{area}(R)$. Any bounded figure Ω can be approximated as the union of N disjoint, arbitrarily small rectangles R_i. Because T is a linear transformation, $\operatorname{area}(\Omega) \approx \sum_{i=1}^{N} \operatorname{area}(R_i)$. Therefore,

$$\operatorname{area}(T(\Omega)) \approx \sum_{i=1}^{N} \operatorname{area}(T(R_i)) = \sum_{i=1}^{N} |\det(\mu(T))| \operatorname{area}(R_i).$$

Factoring out the constant scalar $|\det(\mu(T))|$ yields

$$\operatorname{area}(T(\Omega)) \approx |\det(\mu(T))| \sum_{i=1}^{N} \operatorname{area}(R_i) = |\det(\mu(T))| \operatorname{area}(\Omega).$$

Taking the limit as the size of the rectangles approaches zero, $\operatorname{area}(T(\Omega)) = |\det(\mu(T))| \operatorname{area}(\Omega)$. ⌑

4

Homomorphisms & Quotient Groups

Definitions

Definition 4.1 (Group Homomorphism)**.** If $(G, *)$ and $(H, \circ)$ are groups, then a function $f : G \to H$ is a *homomorphism* if, for all $x, y \in G$, $f(x * y) = f(x) \circ f(y)$. If f is also a bijection then f is called an *isomorphism*. G and H are called *isomorphic*, denoted by $G \cong H$, if there exists an isomorphism between them.

Definition 4.2. If $f : G \to H$ is a homomorphism, define

$$\text{kernel } f = \{x \in G : f(x) = \mathbb{1}\}$$

and

$$\text{image } f = \{h \in H : h = f(x) \text{ for some } x \in G\}.$$

We usually abbreviate kernel f to ker f and image f to im f.

Definition 4.3 (Normal Subgroup)**.** A subgroup K of a group G is called a *normal subgroup* if $k \in K$ and $g \in G$ implies $gkg^{-1} \in K$. If K is a normal subgroup of G, one writes $K \lhd G$.

Remark 4.4. It follows immediately from Theorem 4.10 that the kernel of a homomorphism $f : G \to H$ is always a normal subgroup $\ker f \lhd G$.

Remark 4.5. If G is an abelian group, then every subgroup K is normal, since for all $k \in K$ and $g \in G$, $gkg^{-1} = kgg^{-1} = k \in K$.

Definition 4.6 (Subset Operation)**.** Let $\mathscr{S}(G)$ be the family of all non-empty subsets of a group G. For some $X, Y \in \mathscr{S}(G)$, define the operation $XY = \{xy : x \in X, y \in Y\}$.

Definition 4.7 (Direct Product)**.** If H and K are groups, then their *direct product*, denoted by $H \times K$, is the set $\{(h, k) : h \in H, k \in K\}$ equipped with the operation $(h, k)(h', k') = (hh', kk')$.

Remark 4.8. It is easy to check that $H \times K$ is a group. Inductively, we can see that $H_1 \times H_2 \times \cdots \times H_n$ is also a group.

Definition 4.9 (Quotient Group)**.** Let K be a subgroup of a group G, and let G/K be the family of all cosets of K in G. If K is a normal subgroup, G/K is called the *quotient group* G mod K. This definition follows from Remark 4.13.

Theorems

Theorem 4.10. *Let $f : G \to H$ be a homomorphism. The following statements are true:*

(i) $\ker f$ *is a subgroup of* G *and* $\operatorname{im} f$ *is a subgroup of* H.
(ii) *If* $x \in \ker f$ *and if* $a \in G$*, then* $axa^{-1} \in \ker f$.
(iii) f *is an injection if and only if* $\ker f = \mathbb{1}$.

Proof. Here are the proofs:

(i) Every conclusion follows from Lemma 4.21. First, notice that $\mathbb{1} \in \ker f$ and $\mathbb{1} \in \operatorname{im} f$ since $f(\mathbb{1}) = \mathbb{1}$. To see that $\ker f$ is a subgroup of G, if $x, y \in \ker f$, then $f(x)f(y) = f(xy) = \mathbb{1}$, so $xy \in \ker f$. Finally, if $x \in \ker f$, then $f(x) = \mathbb{1}$, so $f(x^{-1}) = f(x)^{-1} = \mathbb{1}$, and $x^{-1} \in \ker f$. Thus, $\ker f$ is a subgroup of G. Let us now show that $\operatorname{im} f$ is a subgroup of H. If $h, k \in \operatorname{im} f$, $h = f(x)$ and $k = f(y)$ for some $x, y \in G$. Then, $hk = f(x)f(y) = f(xy) \in H$. Finally, if $h = f(x) \in \operatorname{im} f$, then $h^{-1} = f(x)^{-1} = f(x^{-1}) \in \operatorname{im} f$. Thus, $\operatorname{im} f$ is a subgroup of H.

(ii) If $x \in \ker f$, $f(x) = \mathbb{1}$, so $f(axa^{-1}) = f(a)f(x)f(a^{-1}) = f(a)f(a)^{-1} = \mathbb{1}$, so $axa^{-1} \in \ker f$.

(iii) If f is an injection, $x \neq 1$ implies $f(x) \neq f(\mathbb{1}) = \mathbb{1}$, and so $x \notin \ker f$. Conversely, if $\ker f = \mathbb{1}$, assume $f(x) = f(y)$. Then, $f(x)f(y)^{-1} = f(xy^{-1}) = \mathbb{1}$. It follows that $xy^{-1} \in \ker f$, so $xy^{-1} = \mathbb{1}$, and $x = y$.

And we are done. ⌑

Theorem 4.11. *Let G be a group and H be a subgroup. H is a normal subgroup of G if and only if $aH = Ha$ for all $a \in G$.*

Proof. $\boxed{\Longrightarrow}$ Suppose H is a normal subgroup of G. To show that $aH \subset Ha$, let $ah \in aH$ be arbitrary. We show there exists $ka \in Ha$ such that $ah = ka$. Let $k = aha^{-1}$, and the result follows immediately. To show $Ha \subset aH$, let $ka \in Ha$ be arbitrary. We show there exists $ah \in aH$ such that $ka = ah$. Let $h = a^{-1}ka$, and the result follows. This yields $aH = Ha$, which is what we want.

$\boxed{\Longleftarrow}$ Suppose $aH = Ha$ for all $a \in G$. Let $x \in G$ and $h \in H$ be arbitrary. Then, $xh = kx$ for some $k \in H$. Therefore, $xhx^{-1} = k \in H$, so $H \lhd G$, and the proof is complete. ⌑

Theorem 4.12. *Let K be a subgroup of a group G, and let G/K be the family of all the cosets of K in G. If K is a normal subgroup, then $aKbK = abK$ for all $a, b \in G$, and G/K is a group under this operation.*

Proof. Since K is normal, $aK = Ka$ for all $a \in G$. Associativity of the operation in $\mathscr{S}(G)$ gives $(aKbK) = a(Kb)K = a(bK)K = abKK = abK$ since $KK = K$. Thus, the product of two cosets of K is again a coset of K, and so an operation on G/K has been defined. Equality $(XY)Z = X(YZ)$ holds when X, Y, Z are cosets of K, so the operation is associative. The identity is the coset $K = \mathbb{1}K$ since $(\mathbb{1}K)(bK) = bK = (bK)(\mathbb{1}K)$ for all $b \in G$. The inverse of aK is $a^{-1}K$ since $aKa^{-1}K = K = a^{-1}KaK$. Therefore, G/K is a group. ⌑

Remark 4.13. The group G/K is called the *quotient group* G mod K. When G is finite, its order $|G/K|$ is the index $[G : K] = |G|/|K|$.

Theorem 4.14 (First Isomorphism Theorem)**.** *If $f : G \to H$ is a homomorphism with* $\ker f = K$*, then $K \lhd G$ and the function $\varphi : G/K \to$* $\operatorname{im} f \subset H$*, given by $aK \mapsto f(a)$, is an isomorphism.*

Proof. By Remark 4.4, $K \lhd G$. We have $\varphi(a)\varphi(b) = f(a)f(b) = f(ab) = \varphi(ab)$ since f is a homomorphism, so φ is also a homomorphism. It is clear that $\operatorname{im}\varphi = \operatorname{im} f$, so what remains is to show φ is injective. We have

$$\ker\varphi = \{aK \in G/K : \varphi(aK) = \mathbb{1}\} = \{aK \in G/K : f(a) = \mathbb{1}\} = \{aK \in G/K : a \in K\} = \{K\}$$

since $a \in K$ implies $aK = K$ by Lemma 2.21. Hence, $\ker\varphi$ consists of the single element K, which is the identity element, so φ is injective by Theorem 4.10. ⌑

Theorem 4.15. *If m and n are relatively prime, then $\mathbb{Z}_{mn} \cong \mathbb{Z}_m \times \mathbb{Z}_n$.*

Proof. If $a \in \mathbb{Z}$, denote its congruence class in $\mathbb{Z}_m$ by $[a]_m$. Let $f : \mathbb{Z} \to \mathbb{Z}_m \times \mathbb{Z}_n$ be given by $a \mapsto ([a]_m, [a]_n)$. This is a homomorphism since

$$f(a+b) = ([a+b]_m, [a+b]_n) = ([a]_m, [a]_n) + ([b]_m, [b]_n) = f(a) + f(b).$$

Now, we show $\ker f = \langle mn \rangle$, which, because $\mathbb{Z}$ is the additive group, is $\{kmn : k \in \mathbb{Z}\}$. First, it is obvious that $\langle mn \rangle \subset \ker f$, since $f(kmn) = ([0]_m, [0]_n)$ for all k. It remains to show that $\ker f \subset \langle mn \rangle$, so let $x \in \ker f$ be arbitrary. Since $f(x) = ([x]_m, [x]_n) = ([0]_m, [0]_n)$, $m \mid x$ and $n \mid x$. Because $\gcd(m, n) = 1$, it follows that $mn \mid x$, and so $x \in \langle mn \rangle$, which implies $\ker f \subset \langle mn \rangle$; hence, $\ker f = \langle mn \rangle$. We now show f is surjective, so let $([a]_m, [b]_n) \in \mathbb{Z}_m \times \mathbb{Z}_n$. Since $\gcd(m, n) = 1$, the Chinese Remainder Theorem provides a solution $x \in \mathbb{Z}$ for the system

$$\begin{cases} x \equiv a \pmod{m} \\ x \equiv b \pmod{n} \end{cases}$$

Therefore, f is surjective, so $\operatorname{im} f = \mathbb{Z}_m \times \mathbb{Z}_n$. By the First Isomorphism Theorem, we conclude $\mathbb{Z}/\langle mn \rangle = \mathbb{Z}_{mn} \cong \mathbb{Z}_m \times \mathbb{Z}_n$, and the proof is done. ⌑

Theorem 4.16 (Primary Decomposition)**.** *Let G be a finite abelian group of order $|G| = n$. If $n = p_1^{e_1} p_2^{e_2} \dots p_r^{e_r}$ is the unique prime factorization of n, then G is isomorphic to a direct product $H_1 \times H_2 \times \cdots \times H_r$, where every element of H_i has order some power of p_i.*

Proof. Define $H_i = \{g \in G : \operatorname{ord}(g) = p_i^k \text{ for some } k\}$. Using the fact that G is abelian, it is easy to show that each H_i is a subgroup of G. Denote $p_i^{e_i}$ by q_i, so that $n = |G| = q_1 \ldots q_r$, and let us write the abelian group G additively.

We claim that if $g \in G$, then $g = \sum h_i$, where $h_i \in H_i$. Define integers $s_i = n/q_i$. Since $p_i \nmid s_i$ for all i, it follows easily that $\{s_1, \ldots, s_r\}$ is relatively prime; that is the only positive common divisor of the s_i is 1. Since each common divisor can be written as a linear combination of the terms, $1 = \sum t_i s_i$ for some $t_1, \ldots, t_r$. Therefore, $g = \sum t_i s_i g$. Now $q_i(t_i s_i g) = q_i(t_i n/q)g = t_i n g = 0$, by Lagrange's theorem, so that $t_i s_i g \in H_i$, as desired.

Next, we show that this expression for g is unique. Suppose that $g = \sum h_i = \sum k_i$, where $h_i, k_i \in H_i$, so that $0 = \sum (h_i - k_i)$. For each j, $h_j - k_j = \sum_{i \neq j}(h_i - k_i) \in H_i \cap \left\langle H_j : j \neq i \right\rangle$. It suffices, therefore, to show that $H_i \cap \left\langle H_j : j \neq i \right\rangle = \{0\}$. If $x \in H_i$, then $p_i^{e_i} x = 0$; if $x \in \left\langle H_j : j \neq i \right\rangle$, then $mx = 0$, where the only prime factors of m are p_j for $j \neq i$. Therefore, $p_i^{e_i}$ and m are relatively prime, so there are integers s and t with $1 = sp_i^{e_i} + tm$. Hence, $x = sp_i^{e_i} x + tmx = 0 + 0 = 0$ as desired.

Define $\varphi : G \to H_1 \times \cdots \times H_r$ by $\varphi(g) = (h_1, \ldots, h_r)$ where $g = \sum h_i$ and $h_i \in H_i$ for all i. Since such an expression for g is unique, φ is well-defined. It is easy to show that φ is an isomorphism, so $G \cong H_1 \times \cdots \times H_r$, and the proof is complete. ⌑

Theorem 4.17. *Let G be an abelian group of order n. If, for every prime divisor p of n, G contains a unique subgroup of order p, then G is cyclic.*

Proof. Ler $n = p_1^{e_1} p_2^{e_2} \ldots p_\ell^{e_\ell}$ be the unique prime factorization for n, we show this by induction on $\ell \geq 1$. The base case $\ell = 1$ is Lemma 4.23. For the inductive step, let us write $p_1^{e_1} = p^e$ and $p_2^{e_2} \ldots p_\ell^{e_\ell} = q$; of course, $\gcd(p^e, q) = 1$. Now Primary Decomposition gives $G \cong A \times B$ where $|A| = p^e$ and $|B| = q$. By Lemma 4.23, A is cyclic, say with generator a; by induction B is also cyclic, say with generator b. Finally, Lemma 4.22 shows ab has order $p^e q = n$, and so G is cyclic with generator ab, finishing the proof. ⌑

Theorem 4.18 (Basis Theorem)**.** *Every finite abelian group G is isomorphic to a direct product of cyclic groups of prime power orders.*

Proof. By the Primary Decomposition, $G \cong H_1 \times H_2 \times \cdots \times H_r$, where each $|H_i|$ is a prime power. Therefore we may assume that $|G|$ is a power of a prime p; otherwise, we can decompose G to get the same result. By Corollary 2.26, every element in G has order some power of p; choose $h \in G$ or largest order, say $\operatorname{ord}(h) = p^m$, and let $H = \langle h \rangle$. We are going to prove, by complete induction on $|G|$, that G is a direct product of cyclic groups.

The base step $|G| = 1$ is trivially true, since $G = \{\mathbb{1}\}$ os obviously cyclic. Since G is abelian, H is a normal subgroup of G, and the quotient group G/H is defined. Since G/H has order $|G|/|H| < |G|$, the inductive hypothesis states it can be decomposed into a direct product of cyclic groups; that is

$$G/H = \langle x_1 + H \rangle \times \langle x_2 + H \rangle \times \cdots \times \langle x_\ell + H \rangle ,$$

where $x_i + H$ has order p^{e_i} in G/H. By Lemma 4.17, there exists $h_i \in H$ with $p^{e_i}x_i = p^{e_i}h_i$. Define $y_i = x_i - h_i$; note that $y_i + H = x_i + H$ and that $p^{e_i}y_i = 0$. Define $K = \langle y_1, y_2, \ldots, y_\ell \rangle$; as $|K| < |G|$, the inductive hypothesis gives K a direct product of cyclic groups. We are going to prove that $G \cong H \times K$ using Exercise 4.12 (iii), since all subgroups of an abelian group are normal, it suffices to show that $H \cap K = 0$ and $H + K = G$.

It is clear that $0 \in H \cap K$, so we show $H \cap K \subset \{0\}$. If $z \in H \cap K$, then $z \in H$ implies $z + H = 0$ in G/H, while $z \in K$ implies $z = \sum n_i y_i$. Therefore, in G/H, $0 = \sum n_i(y_i + H) = \sum n_i(x_i + H)$. It follows that each term $n_i(x_i + H) = 0$ in G/H, for if an $\ell-$tuple is 0, then each of its coordinates is 0. It follows that $p^{e_i} \mid n_i$, so that $n_i y_i = 0$ for each i because $p^{e_i}y_i = 0$. Hence $x = \sum n_i y_i = 0$. Finally, since $H + K \subset G$ is obvious, we show $G \subset H + K$; if $g \in G$, then $g + H = \sum u_i(y_i + H)$, so that $g - \sum u_i y_i \in H$ and $g \in K + H$, as desired. ⌑

Theorem 4.19 (Second Isomorphism Theorem)**.** *Let H and K be subgroups of a group G with $H \lhd G$. Then, $(H \cap K) \lhd K$ and $K/(H \cap K) \cong HK/H$.*

Proof. The proof is given in Exercise 4.8 (iii). ⌑

Theorem 4.20 (Third Isomorphism Theorem)**.** *Let H and K be normal subgroups of a group G with $K \subset H$. Then, $H/K \lhd G/K$ and $(G/K)/(H/K) \cong G/H$.*

Proof. The proof is given in Exercise 4.9 (ii). ⌑

Lemmas

Lemma 4.21. *Let $f : G \to H$ be a homomorphism of groups. The following statements are true:*

(i) $f(\mathbb{1}) = \mathbb{1}$.
(ii) $f(x^{-1}) = f(x)^{-1}$.
(iii) $f(x^n) = f(x)^n$ *for all* $n \in \mathbb{Z}$.

Proof. The proofs are trivial and left to the reader. ⌑

Lemma 4.22. *Let G be a group, and let $a, b \in G$ commute. If a has order m, b has order n, and* $\gcd(m, n) = 1$*, then ab has order mn.*

Proof. Here we provide a proof for a more generalized version of Lemma 2.22. Since a, b commute, by the Laws of Exponents, $(ab)^r = a^r b^r$ for all r. It follows that $(ab)^{mn} = a^{mn}b^{mn} = \mathbb{1}$, and so it suffices to show that if $(ab)^k = \mathbb{1}$, then $mn \mid k$. If $\mathbb{1} = (ab)^k = a^k b^k$, then $b^{-k} = a^k$. Raising both sides to the power of n yields $(b^n)^{-k} = a^{kn}$, which simplifies to $\mathbb{1} = a^{kn}$ since $\operatorname{ord}(b) = n$. Because $\operatorname{ord}(a) = m$, this implies $m \mid kn$. Since $\gcd(m, n) = 1$, $m \mid kn$ implies $m \mid k$. A similar argument gives $n \mid k$, and so $mn \mid k$. Therefore, for all k such that $(ab)^k = \mathbb{1}$, k is a multiple of mn, so $\operatorname{ord}(ab) = \min\{k \in \mathbb{N}^* : (ab)^k = \mathbb{1}\} = mn$, and the proof is complete. ⌑

Lemma 4.23. *If G is an abelian group of order p^n which has a unique subgroup of order p, then G is cyclic.*

Proof. Choose $a \in G$ of highest order, say, p^m. If $\langle a \rangle = G$, we are done, so suppose, for contradiction, that there exists $b \in G$ such that $b \notin \langle a \rangle$. Since $b^{p^k} = \mathbb{1} \in \langle a \rangle$ for some k, there exists a smallest r such that $b^{p^r} \in \langle a \rangle$. Without loss of generality, replace b with $b^{p^{r-1}}$, in which case $b^p \in \langle a \rangle$. This yields $b^p = a^t$ for some $t \in \mathbb{Z}$, and by factoring t, we have $b^p = a^{up^i}$ for some $u \in \mathbb{Z}$ with $p \nmid u$ and $i \leq m$. If $i = 0$ then $b^p = a^u$, so $b^{p^m} = a^{up^{m-1}} \neq \mathbb{1}$ since $p^m \nmid up^{m-1}$. This means b has a higher order than a, contradicting a having the largest power in G. If $i > 0$, since G is abelian, by the Laws of Exponents, $a^{up^i} b^{-p} = (b^{-1} a^{up^{i-1}})^p = \mathbb{1}$; it follows from an easy contradiction that $\mathrm{ord}(b^{-1} a^{up^{i-1}}) = p$. Hence, it lies in the unique subgroup of order p, namely $\left\langle a^{p^{m-1}} \right\rangle$. Thus, $b^{-1} a^{up^{i-1}} \in \langle a \rangle$, so $b^{-1} \in \langle a \rangle$, implying $b \in \langle a \rangle$, a contradiction. Therefore, no such b exists, and $G = \langle a \rangle$ is cyclic. ⌑

Lemma 4.24. *Let G be a finite abelian group of order p^n, where p is prime, and let $h \in G$ have largest order, say, p^m. If $x \in G$ and $rx \in \langle h \rangle$ for some integer r, then there is $h' \in \langle h \rangle$ with $rx = rh'$.*

Proof. If $p \nmid r$, $\gcd(p, r) = 1$, and Corollary 2.26 says x has order some power of p, say p^g, so that $\gcd(p^g, r) = 1$. Therefore, 1 is a linear combination of p^g and r, say, $1 = sp^g + tr$, and so $x = xsp^g + xtr$ (for $xp^g \neq 0$). But since $p^g = \mathrm{ord}(x)$, $p^g x = 0$, the identity of the group, so if $rx \in \langle h \rangle$, then $x = trx \in \langle h \rangle$, and so we set $h' = tx \in \langle h \rangle$. If $r = up^e$ for some $p \nmid u$, by the previous case, $rx = u(p^e x) \in \langle h \rangle$ implies $p^e x \in \langle h \rangle$, and so we may assume $u = 1$. The hypothesis is thus $p^e x = vp^f h$ where $p \nmid v$. We claim that $e \leq f$; otherwise, $e > f$ and $m - e < m - f$ (where p^m is order of h). Now, $p^m x = p^{m-e}(p^e x) = p^{m-e} v p^f h = v p^{m-e+f} h$. But $v p^{m-e+f} h \neq o$ since h has order p^m while $m - e + f < m$. Therefore, the order of x is larger than the order of h, a contradiction. Since $e \leq f$, we have $p^e x = v p^f h = v p^e p^{f-e} h$, and we set $h' = v p^{f-e} h$. ⌑

Corollaries

Corollary 4.25. *Every normal subgroup K of a group G is the kernel of some homomorphism.*

Proof. We define the *natural map* $\pi : G \to G/K$ by $\pi(a) = aK$, then the formula $aKbK = abK$ can be written as $\pi(a)\pi(b) = \pi(ab)$. Thus, π is a surjective homomorphism. Since K is the identity element in G/K, $\ker \pi = \{a \in G : \pi(a) = K\} = \{a \in G : aK = K\}$, which is exactly K. ⌑

Corollary 4.26. *The Euler's totient function is multiplicative.*

Proof. Quotient groups provide us with yet another proof that the Euler's totient function is multiplicative. By the First Isomorphism Theorem and the proof of Theorem 4.15, when m, n are relatively prime, the function $f : \mathbb{Z}_{mn} \to \mathbb{Z}_m \times \mathbb{Z}_n$ given by $[a] \mapsto ([a]_m, [a]_n)$ is an isomorphism. We will first show that $f(\mathbb{Z}_{mn}^\times) = \mathbb{Z}_m^\times \times \mathbb{Z}_n^\times$. To show $f(\mathbb{Z}_{mn}^\times) \subset \mathbb{Z}_m^\times \times \mathbb{Z}_n^\times$, let $[a] \in \mathbb{Z}_{mn}^\times$. Then there exists $[b] \in \mathbb{Z}_{mn}^\times$ such that $ab \equiv 1 \pmod{mn}$, or $[a][b] = [1] \in \mathbb{Z}_{mn}^\times$. We have

$$f([ab]) = ([ab]_m, [ab]_n) = ([a]_m, [a]_n)([b]_m, [b]_n) = f([a])f([b]) = ([1]_m, [1]_n).$$

This implies $[1]_m = [a]_m[b]_m$ and $[1]_n = [a]_n[b]_n$, so that $f([a]) = ([a]_m, [a]_n) \in \mathbb{Z}_m^\times \times \mathbb{Z}_n^\times$. For the reverse inclusion, if $([c]_m, [e]_n) \in \mathbb{Z}_m^\times \times \mathbb{Z}_n^\times$, then we must show that the unique element $[c] \in \mathbb{Z}_{mn}$ satisfying $f([c]) = ([c]_m, [e]_n)$ belongs to $\mathbb{Z}_{mn}^\times$. There is $[d]_m \in \mathbb{Z}_m$ with $[c]_m[d]_m = [1]_m$ and $[k]_n \in \mathbb{Z}_n$ with $[e]_n[k]_n = [1]_n$. Since f is surjective, there is $b \in \mathbb{Z}$ with $([b]_m, [b]_n) = ([d]_m, [k]_n)$ such that

$$\begin{aligned} f([1]) &= ([1]_m, [1]_n) = ([c]_m[b]_m, [e]_n[b]_n) \\ &= ([c]_m, [e]_n)([b]_m, [b]_n) = f([c])f([b]) \\ &= f([c][b]). \end{aligned}$$

Since f is an injection, $[1] = [c][b]$ and $[c] \in \mathbb{Z}_{mn}^\times$. Therefore, $f(\mathbb{Z}_{mn}^\times) = \mathbb{Z}_m^\times \times \mathbb{Z}_n^\times$; it follows immediately that when $\gcd(m, n) = 1$,

$$\phi(mn) = |\mathbb{Z}_{mn}^\times| = |f(\mathbb{Z}_{mn}^\times)| = |\mathbb{Z}_m^\times \times \mathbb{Z}_n^\times| = |\mathbb{Z}_m^\times| \cdot |\mathbb{Z}_n^\times| = \phi(m)\phi(n).$$

This means the Euler's totient function is multiplicative, and the proof is complete. ⌑

Exercises

4.1. If there is a bijection $f : X \to Y$, then there is an isomorphism $\varphi : S_X \to S_Y$.

Proof. Let $\varphi : S_X \to S_Y$ be defined as $\varphi(\alpha) = f \circ \alpha \circ f^{-1}$. Since $f^{-1} : Y \to X$, $f : X \to Y$, and $\alpha : X \to X$, $\varphi(\alpha) : Y \to Y$. Since compositions of bijections are bijective, $\varphi(\alpha)$ is a bijection, so $\varphi(\alpha) \in S_Y$. Let $\alpha, \beta \in S_X$, we have $\varphi(\alpha)\varphi(\beta) = f \circ \alpha \circ f^{-1} \circ f \circ \beta \circ f^{-1} = f \circ \alpha\beta \circ f^{-1} = \varphi(\alpha\beta)$, which implies φ is a homomorphism. To show that φ is bijective, it can be easily proven that $\varphi^{-1} : S_Y \to S_X$, defined as $\varphi^{-1}(\beta) = f^{-1} \circ \beta \circ f$ is the inverse. Being a bijective homomorphism, φ is an isomorphism, and the proof is complete. ⌑

4.2. Prove that a group G is abelian if and only if the function $f : G \to G$, given by $f(a) = a^{-1}$, is a homomorphism.

Proof. If G is abelian, $f(a)f(b) = a^{-1}b^{-1} = (ab)^{-1} = f(ab)$, so f is a homomorphism. Conversely, if f is a homomorphism, $f(a)f(b) = a^{-1}b^{-1} = f(ab) = (ab)^{-1}$. However, $a^{-1}b^{-1} = (ba)^{-1}$, so $(ab)^{-1} = (ba)^{-1}$, which implies $ab = ba$, and G is abelian. ⌑

4.3. Let $f : G \to H$ be an isomorphism.

(i) Show that if $a, b \in G$ commute, then $f(a), f(b) \in H$ commute. Conclude that if G is abelian, then H is abelian.

(ii) If $a \in G$ has order n, then $f(a) \in H$ has order n. Conclude that if G has an element of some order n and H does not, then $G \not\cong H$.

(iii) Prove that if $G \cong H$, then, for every divisor k of $|G|$, both G and H have the same number of elements of order k.

Proof. Here are the proofs:

(i) If $a, b \in G$ commute, then $f(a)f(b) = f(ab) = f(ba) = f(b)f(a)$, so $f(a), f(b) \in H$ commute. If G is abelian, for all $h, k \in H$, $f^{-1}(h), f^{-1}(k) \in G$ commute (we know the inverse exists since f is bijective), which means h, k also commute. This implies H is abelian.

(ii) Let $\text{ord}(a) = n$, we have $a^n = \mathbb{1}$ implies $f(a^n) = f(a)^n = \mathbb{1}$. Therefore, we know the order of $f(a)$, call it k, must be a divisor of n, which implies $k \leq n$. Conversely, since $k = \text{ord}(f(a))$, we have $f(a)^k = \mathbb{1}$. Because f is a homomorphism, $f(a^k) = \mathbb{1} = f(\mathbb{1})$. Since f is an isomorphism, it is injective, which implies $a^k = \mathbb{1}$. Because n is the smallest positive integer such that $a^n = \mathbb{1}$, it must be that $n \leq k$. The final conclusion follows immediately as the contrapositive of this result.

(iii) Let $O_k(G)$ be the set of elements in G with order k, and $O_k(H)$ be the set of elements in H with order k. We have shown that for each $a \in O_k(G)$, $f(a) \in O_k(H)$, so $|O_k(H)| \geq |O_k(G)|$. Notice $f^{-1} : H \to G$ is also an isomorphism, so the same argument can be applied to show that $|O_k(G)| \geq |O_k(H)|$. This yields $|O_k(G)| = |O_k(H)|$, and the proof is complete.

And we are done. ⌑

4.4. Show that every group G with $|G| < 6$ is abelian.

Proof. Let G be a group and $k = |G|$ be its order. If $k = 1$, $G = \{\mathbb{1}\}$, which is obviously abelian. If $k \in \{2, 3, 5\}$, since k is prime, by Corollary 2.27, G is cyclic. It follows by Corollary 2.28 that G is abelian. When $k = 4$, by Exercise 2.4, G is also abelian. This completes the proof. ⌑

4.5. Let G be the additive group of all polynomials in x with coefficients in $\mathbb{Z}$, and let H be the multiplicative group of all positive rationals. Prove that $G \cong H$.

Proof. Define p_k as the $(k+1)$-st prime number. Let $\varphi : G \to H$ be defined as

$$\varphi(e_0 + e_1 x + \cdots + e_n x^n) = p_0^{e_0} p_1^{e_1} \ldots p_n^{e_n};$$

we show that this is an isomorphism. First, let $e_0 + e_1 x + \cdots + e_n x^n$ and $c_0 + c_1 x + \cdots + c_m x^m$ be two polynomials. Without loss of generality, assume $n \leq m$. We have

$$\varphi(e_0 + e_1 x + \cdots + e_n x^n)\varphi(c_0 + c_1 x + \cdots + c_m x^m) = p_0^{e_0} p_1^{e_1} \ldots p_n^{e_n} p_0^{c_0} p_1^{c_1} \ldots p_m^{c_m}.$$

Since the multiplicative group H is commutative,

$$p_0^{e_0} p_1^{e_1} \ldots p_n^{e_n} p_0^{e_0} p_1^{e_1} \ldots p_m^{e_m} = p_0^{e_0+c_0} p_1^{e_1+c_1} \ldots p_n^{e_n+c_n} \ldots p_m^{c_m}.$$

This is exactly $\varphi((e_0+c_0)+(e_1+c_1)x+\cdots+(e_n+c_n)x^n+\cdots+c_mx^m)$. Therefore, φ is a homomorphism. To show φ is injective, suppose $\varphi(P(x)) = \varphi(C(x))$. This implies $p_0^{e_0}p_1^{e_1}\dots p_m^{e_m} = p_0^{c_0}p_1^{c_1}\dots p_m^{c_m}$. We know that the prime factorization of any positive rational number is unique. Therefore, we must have $e_i = c_i$ for all i, which means $P(x) = C(x)$. Thus, φ is injective. To show φ is surjective, let $q \in H$ be any positive rational number. By the prime factorization of rational numbers, q can be uniquely written as a finite product of prime powers $q = p_0^{r_0}p_1^{r_1}\dots p_k^{r_k}$ where $r_i \in \mathbb{Z}$. The polynomial $R(x) = r_0 + r_1x + \cdots + r_kx^k$ belongs to G, and clearly $\varphi(R(x)) = q$. Thus, φ is surjective. Being a bijective homomorphism, φ is an isomorphism, and $G \cong H$. ⌑

4.6. Prove that a subgroup H of G with $[G : H] = 2$ is a normal subgroup. Conclude that if $[G : H] = 2$, $g^2 \in H$ for all $g \in G$.

Proof. First, to show a subgroup with index 2 is normal, let $a \in G$ be arbitrary. If $a \in H$, $aH = H = Ha$. If $a \notin H$, $aH \neq H$, so $aH = G \setminus H$. Likewise, $Ha \neq H$ so $Ha = G \setminus H = aH$. Since $aH = Ha$ for all $a \in G$, by Theorem 4.11, $H \lhd G$. To show $g^2 \in H$ for all $g \in G$, let $g \in G$ be arbitrary. If $g \in H$, it follows directly that $g^2 \in H$. If $g \notin H$, then as shown above, $gH = G \setminus H$. Suppose, for contradiction, that $g^2 \notin H$. Then $g^2 \in gH$, so $g^2 = gh$ for some $h \in H$, which implies $g = h \in H$, a contradiction. Therefore, $g^2 \in H$ for all $g \in G$. ⌑

4.7. Prove that A_n is a normal subgroup of S_n of order $\frac{n!}{2}$. Conclude that A_4 is the only subgroup of S_4 of order 12.

Proof. By Lemma 2.23, we know $|S_n| = n!$. Let $p : A_n \to S_n \setminus A_n$ be defined as $p(\alpha) = \alpha\tau$ for some fixed transposition τ. This is clearly a function. To show p is injective, notice $\alpha\tau = \beta\tau$ implies $\alpha = \beta$. To show p is surjective, let σ be an arbitrary odd permutation. Since $\operatorname{sgn}(\sigma\tau^{-1}) = \operatorname{sgn}(\sigma)\operatorname{sgn}(\tau^{-1}) = 1$, $\sigma\tau^{-1}$ is an even permutation. Therefore, there exists $\mu \in A_n$ such that $\mu = \sigma\tau^{-1}$, so $\sigma = \mu\tau$, showing p is surjective. Since p is bijective, the numbers of even and odd permutations are the same, so $|A_n| = \frac{|S_n|}{2} = \frac{n!}{2}$. It follows that $[S_n : A_n] = 2$, and $A_n \lhd S_n$ by Exercise 4.6. To show A_4 is the only subgroup of S_4 with order 12, notice that $|A_4| = 12$. It remains to show A_4 is unique, so let H be a subgroup of S_4 with order 12. Let $\alpha \in A_4$ be arbitrary, by Theorem 1.21, $\alpha = \sigma_1\sigma_2\dots\sigma_n$ for each σ_i being a 3-cycle. By Exercise 4.6, $\sigma_i^2 \in H$ for all σ_i. But since each 3-cycle has order 3, $\sigma_i^3 = \mathbb{1}$, so $(\sigma_i^2)^2 = \sigma_i^4 = \sigma_i \in H$ by the closure of H. It follows that $\alpha \in H$, which means $A_4 \subset H$. Since $|A_4| = |H| = 12$, it must be that $H = A_4$, and the proof is complete. ⌑

4.8. Let H and K be subgroups of a group G with $H \lhd G$.

(i) Show that $HK = \{hk : h \in H \text{ and } k \in K\}$ is a subgroup of G, and show that H is a normal subgroup of HK.

(ii) Show that every coset in HK/H has the form kH for some $k \in K$.

(iii) Show that $(H \cap K) \lhd K$ and that $K/(H \cap K) \cong HK/H$.

Proof. Here are the proofs:

(i) To show HK is a subgroup of G, first notice $\mathbb{1} = \mathbb{1}\mathbb{1} \in HK$. Let $x_1 = h_1k_1$ and $x_2 = h_2k_2$, we have $x_1x_2 = h_1k_1h_2k_2$, but since $H \lhd G$, $k_1H = Hk_1$, so $k_1h_2 = h_3k_1$ for some $h_3 \in H$, which makes $x_1x_2 = h_1h_3k_1k_2 \in HK$. Finally, let $x = hk$; using the same argument, $x^{-1} = k^{-1}h^{-1} = h_2k^{-1}$ for some $h_2 \in H$, so $x^{-1} \in HK$. To show $H \lhd HK$, notice that since H and HK are both subgroups of G and H is a subset of HK, it is natural that H is a subgroup of HK. Let $h \in H$ and $h_1k_1 \in HK$, we need to show $h_1k_1hk_1^{-1}h_1^{-1} \in H$. Since $H \lhd G$, we can use the same argument, which arrives at $h_1k_1hk_1^{-1}h_1^{-1} = h_1h_2h_2k_1k_1^{-1}$ for some $h_2, h_3 \in H$, but this is $h_1h_2h_3 \in H$.

(ii) For all $A \in HK/H$, $A = hkH$ for some $hk \in HK$. Since $h \in H$, $hH = H$. Since $H \lhd G$, we have $Hk = kH$, which means $hkH = h(Hk) = (hH)k = Hk = kH$.

(iii) Let $f : K \to HK/H$ be defined as $f(k) = kH$. Since $H \lhd G$,

$$f(k_1k_2) = (k_1k_2)H = k_1(k_2H) = k_1(Hk_2) = (k_1H)(k_2H) = f(k_1)f(k_2),$$

so f is a homomorphism. For all $x \in (H \cap K)$, $f(x) = xH = H$ since $x \in H$, so $x \in \ker f$. For all $x \in \ker f \subset K$, $f(x) = xH = H$, so it must be that $x \in H$, which makes $x \in (H \cap K)$. We then have $(H \cap K) = \ker f$, so it is a normal subgroup of K by the First Isomorphism Theorem. It remains to show that $\operatorname{im} f = HK/H$, or f is surjective. Since every coset in HK/H can be expressed as kH for some $k \in K$, for all $A \in HK/H$, there exists $k \in K$ such that $f(k) = A$, so $\operatorname{im} f = HK/H$. It follows from the First Isomorphism Theorem that $K/(H \cap K) \cong HK/H$.

And we are done. ⌑

4.9. Let H and K be normal subgroups of a group G with $K \subset H$.

(i) Show that $f : G/K \to G/H$, given by $aK \mapsto aH$, is a surjective homomorphism.

(ii) Show that $H/K \lhd G/K$ and that $(G/K)/(H/K) \cong G/H$.

Proof. Here are the proofs:

(i) First, to show f is well-defined, let $aK = bK$ for some $a, b \in G$. Then, we have $b^{-1}aK = K$, so $b^{-1}a \in K$. Since $K \subset H$, $b^{-1}a \in H$, so $b^{-1}aH = H$, and $aH = bH$. To show f is surjective, for any $aH \in G/H$, consider $aK \in G/K$; this yields $f(aK) = aH$, which is needed. Let $a_1, a_2 \in G$, since $K \lhd G$, $a_1Ka_2K = a_1(a_2K)K = a_1a_2K$, and

$$f(a_1Ka_2K) = f(a_1a_2K) = (a_1a_2)H = (a_1H)(a_2H) = f(a_1K)f(a_2K),$$

so f is a homomorphism, completing the proof.

(ii) We show that $\ker f = H/K$. First, if $hK \in H/K$, $f(hK) = hH = H$ so $H/K \subset \ker f$. If $f(aK) = aH = H$, $a \in H$, and $aK \in H/K$, so $\ker f \subset H/K$, which means $\ker f = H/K$. By the First Isomorphism Theorem, $H/K \lhd G/K$, and since $\operatorname{im} f = G/H$, $(G/K)/(H/K) \cong G/H$.

And we are done. ◻

4.10. Suppose H and K are groups. Let $H^* = \{(h, \mathbb{1}) : h \in H\}$ and $K^* = \{(\mathbb{1}, k) : k \in K\}$.

(i) Prove that $H^* \lhd (H \times K)$ and $K^* \lhd (H \times K)$ with $H \cong H^*$ and $K \cong K^*$.
(ii) Prove that $(H \times K)/K^* \cong H$.

Proof. Here are the proofs:

(i) To establish the isomorphisms, let $f : H \to H \times K$ be defined by $h \mapsto (h, \mathbb{1})$. For any $h_1, h_2 \in H$, we have $f(h_1h_2) = (h_1h_2, \mathbb{1}) = (h_1, \mathbb{1})(h_2, \mathbb{1}) = f(h_1)f(h_2)$, so f is a homomorphism. Since f is clearly injective and its image is exactly $\operatorname{im} f = H^*$, it restricts to an isomorphism $H \cong H^*$. Because the image of a homomorphism is a subgroup, H^* is a subgroup of $H \times K$. By an identical argument, $g : K \to H \times K$ defined by $k \mapsto (\mathbb{1}, k)$ is a homomorphism with $\operatorname{im} g = K^*$, yielding $K \cong K^*$ and showing K^* is a subgroup of $H \times K$.
Now, let $(h_1, k_1) \in H \times K$ and $(h, \mathbb{1}) \in H^*$ be arbitrary; we show $(h_1, k_1)(h, \mathbb{1})(h_1^{-1}, k_1^{-1}) \in H^*$. Indeed,
$$(h_1, k_1)(h, \mathbb{1})(h_1^{-1}, k_1^{-1}) = (h_1hh_1^{-1}, k_1\mathbb{1}k_1^{-1}) = (h_1hh_1^{-1}, \mathbb{1}) \in H^*,$$
since $h_1hh_1^{-1} \in H$. Thus, $H^* \lhd (H \times K)$. The same argument can be used to conclude $K^* \lhd (H \times K)$.
(ii) Let $\pi : H \times K \to H$ be defined by $(h, k) \mapsto h$. Let $(h_1, k_1), (h_2, k_2) \in H \times K$, we have
$$\pi((h_1, k_1)(h_2, k_2)) = h_1h_2 = \pi(h_1, k_1)\pi(h_2, k_2),$$
so π is a homomorphism. We also see that $\pi(K^*) = \{\mathbb{1}\}$ and $\pi(h, k) = h = \mathbb{1}$ implies $h = \mathbb{1}$ and $(\mathbb{1}, k) \in K^*$, so K^* is exactly $\ker \pi$. Additionally, for all $h \in H$, there exists $(h, \mathbb{1}) \in H \times K$ with $\pi(h, \mathbb{1}) = h$, so $\operatorname{im} \pi = H$. The First Isomorphism Theorem yields $(H \times K)/K^* \cong H$ as desired.

And we are done. ◻

4.11. Let G be a finite group with $K \lhd G$. If $\gcd(|K|, [G : K]) = 1$, prove that K is the unique subgroup of G having order $|K|$.

Proof. Let H be a subgroup of G with $|H| = |K|$. By Exercise 4.8, HK is a subgroup of G. By the Second Isomorphism Theorem, $H/(H \cap K) \cong HK/K$, so $|H/(H \cap K)| = |HK/K|$. This yields $|H|/|(H \cap K)| = |HK|/|K|$ and $|HK| = \frac{|H||K|}{|H \cap K|} = \frac{|K|^2}{|H \cap K|}$. Notice by Lagrange's Theorem that
$$\frac{|K|^2}{|H \cap K|} = |HK| \mid |G| = |K|[G : K],$$
and by canceling $|K|$ on both sides we get $\frac{|K|}{|H \cap K|} \mid [G : K]$. However, since it is clear that $\frac{|K|}{|H \cap K|} \mid |K|$ as well, $\frac{|K|}{|H \cap K|}$ is a common divisor of $|K|$ and $[G : K]$. Because $\gcd(|K|, [G : K]) = 1$, $\frac{|K|}{|H \cap K|} = 1$, so $|K| = |H \cap K|$. This implies $H \cap K = K$, which means $K \subset H$. Since G is a finite group and $|H| = |K|$, it follows that $H = K$, and the proof is complete. ◻

4.12. Let G be a group containing normal subgroups H and K with $H \cap K = \mathbb{1}$ and $HK = G$.

(i) If $g \in G$, then the condition $HK = G$ says that $g = hk$ for some $h \in H$ and $k \in H$. Prove that this factorization of g is unique.
(ii) Prove that if $h \in H$ and $k \in K$, then $hk = kh$.
(iii) Prove that $G \cong H \times K$.

Proof. Here are the proofs:

(i) Suppose $g = hk = h'k'$ for some $h' \in H$, $k' \in K$. Then, we have $h'^{-1}h = k'k^{-1} \in (H \cap K)$. Since $H \cap K = \{\mathbb{1}\}$, $h'^{-1}h = k'k^{-1} = \mathbb{1}$, so $h' = h$ and $k' = k$ as desired.
(ii) Since $H \lhd G$ and $K \lhd G$, $hkh^{-1} \in K$ and $khk^{-1} \in H$. Since $hkh^{-1} \in K$, $k^{-1}hkh^{-1} \in K$. Using the same argument, we get $k^{-1}hkh^{-1} \in H$. Therefore, $k^{-1}hkh^{-1} \in (H \cap K)$, so $k^{-1}hkh^{-1} = \mathbb{1}$, and it follows that $hk = kh$ as desired.
(iii) Define $f : G \to H \times K$ by $g \mapsto (h, k)$ where $g = hk$ is its unique factorization. Let $g_1, g_2 \in G$ where $g_1 = h_1k_1$ and $g_2 = h_2k_2$. We have $g_1g_2 = h_1k_1h_2k_2 = h_1h_2k_1k_2$, so

$$f(g_1g_2) = (h_1h_2, k_1k_2) = (h_1, k_1)(h_2, k_2) = f(g_1)f(g_2),$$

and f is a homomorphism. We can clearly see the inverse is $f^{-1} : H \times K \to G$ defined by $(h, k) \mapsto hk$, so f is bijective, and it is an isomorphism. Therefore, $G \cong H \times K$.

And we are done. ⌑

4.13. Prove the following statements about the Euler's totient function:

(i) If p is a prime, $\phi(p^n) = p^n\left(1 - \frac{1}{p}\right)$.
(ii) Let $h \in \mathbb{Z}$ and $h = p_1^{q_1}p_2^{q_2}\dots p_n^{q_n}$ be its unique prime factorization, then

$$\phi(h) = h\left(1 - \frac{1}{p_1}\right)\left(1 - \frac{1}{p_2}\right)\dots\left(1 - \frac{1}{p_n}\right).$$

Proof. Here are the proofs:

(i) The numbers less than or equal to p^n that are not coprime to p^n are precisely the multiples of p, which are of the form kp for $1 \le k \le p^{n-1}$. There are exactly p^{n-1} such multiples. Therefore, $\phi(p^n) = p^n - p^{n-1} = p^n\left(1 - \frac{1}{p}\right)$ as desired.
(ii) Since each p_i is pairwise coprime, by Corollary 4.26, we have

$$\phi(h) = \phi(p_1^{q_1})\phi(p_2^{q_2})\dots\phi(p_n^{q_n}) = p_1^{q_1}\left(1 - \frac{1}{p_1}\right)\dots p_n^{q_n}\left(1 - \frac{1}{p_n}\right) = h\left(1 - \frac{1}{p_1}\right)\dots\left(1 - \frac{1}{p_n}\right)$$

And we are done. ⌑

4.14. Prove the following statements:

(i) Let $p_1, p_2, \ldots, p_t$ be distinct primes. If a group $L = C_1 \times C_2 \times \cdots \times C_t$, where each C_j is a cyclic group of order some power of p_j, then L is cyclic.

(ii) Every finite abelian group G is a direct product of cyclic groups $G = K_1 \times K_2 \times \cdots \times K_\ell$ with $k_1 \mid k_2 \mid \cdots \mid k_\ell$, where $k_i = |K_i|$ for all i.

Proof. Here are the proofs:

(i) Since for each cyclic C_i, $|C_i| = p_i^{q_i}$, there exists $c_i \in C_i$ with $\mathrm{ord}(c_i) = p_i^{q_i}$. We know that $|L| = p_1^{q_1} p_2^{q_2} \ldots p_t^{q_t}$. Construct $\ell \in L$ where $\ell = (c_1, c_2, \ldots c_t)$. We have

$$\ell^{p_1^{q_1} p_2^{q_2} \ldots p_t^{q_t}} = (c_1^{p_1^{q_1} p_2^{q_2} \ldots p_t^{q_t}}, c_2^{p_1^{q_1} p_2^{q_2} \ldots p_t^{q_t}}, \ldots c_t^{p_1^{q_1} p_2^{q_2} \ldots p_t^{q_t}}) = (\mathbb{1}, \mathbb{1}, \ldots, \mathbb{1}),$$

so it remains to show that this is the smallest number to satisfy this condition. Suppose $\ell^m = \mathbb{1}$, then $(c_1^m, c_2^m, \ldots, c_t^m) = (\mathbb{1}, \mathbb{1}, \ldots, \mathbb{1})$, so $c_i^m = \mathbb{1}$ for every i. Since $\mathrm{ord}(c_i) = p_i^{q_i}$, it follows that $p_i^{q_i} \mid m$ for each i. Because the numbers $p_1^{q_1}, p_2^{q_2}, \ldots, p_t^{q_t}$ are pairwise relatively prime, their product $p_1^{q_1} p_2^{q_2} \cdots p_t^{q_t}$ divides m. Hence $\mathrm{ord}(\ell) = p_1^{q_1} p_2^{q_2} \cdots p_t^{q_t} = |L|$, making L cyclic.

(ii) Let $|G| = p_1^{q_1} p_2^{q_2} \cdots p_n^{q_n}$. By Primary Decomposition, $G \cong H_1 \times H_2 \times \cdots \times H_n$, where each H_i has order a power of p_i. Since each H_i is a finite abelian p_i-group, it can be decomposed as $H_i \cong C_{i1} \times C_{i2} \times \cdots \times C_{ir_i}$, where each C_{ij} is cyclic of order $p_i^{a_{ij}}$ and $a_{i1} \geq a_{i2} \geq \cdots \geq a_{ir_i}$. If necessary, append trivial groups so that every decomposition has the same number ℓ of factors. For each $1 \leq j \leq \ell$, define $K_j = C_{1j} \times C_{2j} \times \cdots \times C_{nj}$. Since the orders of the factors of K_j are powers of distinct primes, they are pairwise relatively prime. Hence each K_j is cyclic. Moreover, $|K_j| = \prod_{i=1}^n p_i^{a_{ij}}$. Since $a_{i1} \geq a_{i2} \geq \cdots \geq a_{i\ell}$ for every prime p_i, it follows that $k_1 \mid k_2 \mid \cdots \mid k_\ell$, where $k_i = |K_i|$ for all i. Finally,

$$G \cong (H_1 \times \cdots \times H_n) \cong (C_{11} \times \cdots \times C_{n1}) \times \cdots \times (C_{1\ell} \times \cdots \times C_{n\ell}) \cong K_1 \times K_2 \times \cdots \times K_\ell.$$

Therefore, $G \cong K_1 \times K_2 \times \cdots \times K_\ell$, where each K_j is cyclic and $k_1 \mid k_2 \mid \cdots \mid k_\ell$, as desired.

And we are done. ⌑

Appendix

Here are some of my personal favorite goodies that are essentially not in any other sections.

Theorem A.1 (Leibniz)**.** *A function $f : \mathbb{R} \to \mathbb{R}$ is call a C^∞-function if it has a k^{th} derivative $D^k f$ for every $k \geq 0$. Prove that if f and g are C^∞-function, then*

$$D^n(fg) = \sum_{k=0}^{n} \binom{n}{k} (D^k f)(D^{n-k} g).$$

Proof. We proceed by induction on $n \geq 0$. For $n = 0$, the left-hand side is $D^0(fg) = fg$. The right-hand side is:

$$\sum_{k=0}^{0} \binom{0}{k} (D^k f)(D^{0-k} g) = \binom{0}{0} (D^0 f)(D^0 g) = fg.$$

Thus, the statement holds for $n = 0$. For the inductive step, assume the statement holds for some $n \geq 0$, i.e.,

$$D^n(fg) = \sum_{k=0}^{n} \binom{n}{k} (D^k f)(D^{n-k} g).$$

We want to show that it holds for $n + 1$. Applying the derivative operator D to both sides and using the standard product rule $D(uv) = (Du)v + u(Dv)$, we get:

$$\begin{aligned} D^{n+1}(fg) &= D\left(\sum_{k=0}^{n} \binom{n}{k} (D^k f)(D^{n-k} g)\right) = \sum_{k=0}^{n} \binom{n}{k} D\left((D^k f)(D^{n-k} g)\right) \\ &= \sum_{k=0}^{n} \binom{n}{k} \left[(D^{k+1} f)(D^{n-k} g) + (D^k f)(D^{n-k+1} g)\right]. \end{aligned}$$

Splitting this into two separate sums yields

$$\sum_{k=0}^{n} \binom{n}{k} (D^{k+1} f)(D^{n-k} g) + \sum_{k=0}^{n} \binom{n}{k} (D^k f)(D^{n-k+1} g).$$

Shift the index of the first sum by letting $j = k + 1$ (so $k = j - 1$), we get

$$\sum_{j=1}^{n+1} \binom{n}{j-1} (D^j f)(D^{n-j+1} g) + \sum_{k=0}^{n} \binom{n}{k} (D^k f)(D^{n-k+1} g).$$

Changing the dummy variable j back to k in the first sum allows us to align the terms to make

$$\sum_{k=1}^{n+1} \binom{n}{k-1} (D^k f)(D^{n-k+1} g) + \sum_{k=0}^{n} \binom{n}{k} (D^k f)(D^{n-k+1} g).$$

Now, pull out the $k = n + 1$ term from the first sum and the $k = 0$ term from the second sum:

$$\binom{n}{0}(D^0 f)(D^{n+1} g) + \sum_{k=1}^{n} \left[\binom{n}{k-1} + \binom{n}{k} \right] (D^k f)(D^{n-k+1} g) + \binom{n}{n}(D^{n+1} f)(D^0 g).$$

Using Pascal's identity and noting that $\binom{n}{0} = \binom{n+1}{0} = 1$ and $\binom{n}{n} = \binom{n+1}{n+1} = 1$, we get

$$\binom{n+1}{0}(D^0 f)(D^{n+1} g) + \sum_{k=1}^{n} \binom{n+1}{k}(D^k f)(D^{n+1-k} g) + \binom{n+1}{n+1}(D^{n+1} f)(D^0 g)$$

which is exactly

$$\sum_{k=0}^{n+1} \binom{n+1}{k}(D^k f)(D^{n+1-k} g).$$

This completes the inductive step. By induction, the formula holds for all $n \geq 0$. ⌑

Theorem A.2 (Fundamental Theorem of Arithmetic)**.** *Assume that an integer $a \geq 2$ has factorizations*

$$a = p_1 p_2 \ldots p_m \quad \text{and} \quad a = q_1 q_2 \ldots q_n,$$

where the p's and q's are primes. Then $n = m$ and the q's may be reindexed so that $q_i = p_i$ for all i.

Proof. Suppose, for contradiction, that there exist some "criminals". Then, by the Well-Ordering Principle, there is a least "criminal", that is, $a = p_1 p_2 \ldots p_m = q_1 q_2 \ldots q_n$, where some factor occurs with different frequencies in the two factorizations, and a is the smallest such "criminal". The equation gives $p_m \mid q_1 \ldots q_n$. Then, there is some i with $p_m \mid q_i$. But q_i, being a prime, has no positive divisors other than 1 and itself, so that $q_i = p_m$. Reindexing, we may assume that $q_n = p_m$. Cancelling, we have $p_1 \ldots p_{m-1} = q_1 \ldots q_{n-1} = a/p_m < a$. Since a is the least criminal, $n - 1 = m - 1$ and the q's maybe reindexed so that $q_i = p_i$ for all i. It follows that a is not a "criminal", contradiction. ⌑

To Be Continued...

References

[1] Joseph J. Rotman. *A First Course in Abstract Algebra*. Prentice Hall, Upper Saddle River, NJ, 1996.